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Methods for enhancing efficacy of therapeutic immune cells

Abstract

The present invention relates to a method of using a receptor (e.g., chimeric antigen receptor—CAR) that activates an immune response upon binding a cancer cell ligand in conjunction with a target-binding molecule that targets a protein or molecule for removal or neutralization to generate enhanced anti-cancer immune cells. The present invention also relates to engineered immune cells having enhanced therapeutic efficacy and uses thereof.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4946778	12/1989	Ladner et al.	N/A	N/A
5260203	12/1992	Ladner et al.	N/A	N/A
7887805	12/2010	Pedersen et al.	N/A	N/A
8119775	12/2011	Moretta	N/A	N/A
8399645	12/2012	Campana et al.	N/A	N/A
8580714	12/2012	Almagro et al.	N/A	N/A
8614307	12/2012	Moretta et al.	N/A	N/A

8637258	12/2013	Padkjaer et al.	N/A	N/A
8796427	12/2013	Spee	N/A	N/A
8981065	12/2014	Moretta et al.	N/A	N/A
9181527	12/2014	Sentman	N/A	N/A
9273283	12/2015	Sentman	N/A	N/A
9422368	12/2015	Spee et al.	N/A	N/A
9683042	12/2016	Lee et al.	N/A	N/A
9902936	12/2017	Moretta et al.	N/A	N/A
10550183	12/2019	Png et al.	N/A	N/A
10730942	12/2019	Pule et al.	N/A	N/A
10765699	12/2019	Campana et al.	N/A	N/A
11161907	12/2020	Powell	N/A	C12N 15/113
11440958	12/2021	Png et al.	N/A	N/A
11648269	12/2022	Campana et al.	N/A	N/A
11679132	12/2022	Campana et al.	N/A	N/A
11945865	12/2023	Png et al.	N/A	N/A
2005/0282181	12/2004	Yan et al.	N/A	N/A
2006/0034834	12/2005	Marasco et al.	N/A	N/A
2007/0036773	12/2006	Cooper et al.	N/A	N/A
2009/0196850	12/2008	Romagne et al.	N/A	N/A
2012/0282256	12/2011	Campana et al.	N/A	N/A
2013/0266551	12/2012	Campana et al.	N/A	N/A
2013/0287748	12/2012	June	N/A	N/A
2014/0120622	12/2013	Gregory	N/A	N/A
2014/0186387	12/2013	Lauer	N/A	N/A
2016/0256488	12/2015	Wu	N/A	N/A
2016/0312182	12/2015	Sentman	N/A	N/A
2017/0335331	12/2016	Zhao et al.	N/A	N/A
2018/0008638	12/2017	Campana	N/A	N/A
2018/0066034	12/2017	Ma et al.	N/A	N/A
2018/0086831	12/2017	Pule	N/A	N/A
2019/0002912	12/2018	Lu et al.	N/A	N/A
2019/0038733	12/2018	Campana et al.	N/A	N/A
2019/0046571	12/2018	Campana et al.	N/A	N/A
2019/0345217	12/2018	Ma et al.	N/A	N/A
2020/0087398	12/2019	Qasim et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2 937 157	12/2017	CA	N/A
107709548	12/2017	CN	N/A
113713091	12/2020	CN	N/A
107709548	12/2021	CN	N/A
115029362	12/2021	CN	N/A
0404097	12/1989	EP	N/A
2247619	12/2009	EP	N/A
3169773	12/2016	EP	N/A
3322801	12/2017	EP	N/A
3359168	12/2017	EP	N/A
3474867	12/2018	EP	N/A
3568467	12/2018	EP	N/A
H03219896	12/1990	JP	N/A
H09501824	12/1996	JP	N/A
2001516766	12/2000	JP	N/A
2004529610	12/2003	JP	N/A
2008506368	12/2007	JP	N/A
2008518021	12/2007	JP	N/A
2009511495	12/2008	JP	N/A
2010537671	12/2009	JP	N/A
2011510047	12/2010	JP	N/A
2014507118	12/2013	JP	N/A
6895380	12/2020	JP	N/A
2021137024	12/2020	JP	N/A
WO-8801649	12/1987	WO	N/A
WO-9311161	12/1992	WO	N/A
WO 99/14353	12/1998	WO	N/A
WO 2003/051926	12/2002	WO	N/A
WO-2005017163	12/2004	WO	N/A
WO 2006/003179	12/2005	WO	N/A
WO 2009/092805	12/2008	WO	N/A
WO 2012/079000	12/2011	WO	N/A
WO-2013126712	12/2012	WO	N/A
WO-2014011984	12/2013	WO	N/A
WO 2014/124143	12/2013	WO	N/A
WO-2014134165	12/2013	WO	N/A

WO-2014159940	12/2013	WO	N/A
WO 2015/075468	12/2014	WO	N/A
WO-2015121454	12/2014	WO	N/A
WO 2015/150771	12/2014	WO	N/A
WO 2016/055551	12/2015	WO	N/A
WO 2016/102965	12/2015	WO	N/A
WO 2016/126213	12/2015	WO	N/A
WO 2017/213979	12/2016	WO	N/A
WO 2018/027036	12/2017	WO	N/A
WO-2019032916	12/2018	WO	N/A

OTHER PUBLICATIONS

Lanitis et al A Human ErbB2-Specific T-Cell Receptor Confers Potent Antitumor Effector Functions in Genetically Engineered Primary Cytotoxic Lymphocytes. *Journal of Immunology*, 176(1):100-107 (2006). cited by applicant

Alanen et al., Beyond KDEL: the role of positions 5 and 6 in determining ER localization. *J. Mol. Biol.* (2011) 409, 291-297. Available online Apr. 6, 2011. cited by applicant

Alarcon et al., Assembly of the human T cell receptor-CD3 complex takes place in the endoplasmic reticulum and involves intermediary complexes between the alpha and beta chains, *Journal of Biological Chemistry*, Feb. 25, 1988;263(6):2953-61. cited by applicant

Almagro, Juan C., Fransson, Johan. Humanization of antibodies. *Frontiers in Bioscience* 13;1619-1633 (Jan. 1, 2008). cited by applicant

Altschul et al. Basic local alignment search tool. *J Mol Biol* 215(3):403-410 (1990). cited by applicant

Anti-CD3 epsilon [OKT-3 (muromonab)]. Absolute Antibody. Website. Copyright 2021. Retrieved Dec. 26, 2021 at URL: <https://absoluteantibody.com/products/anti-cd3epsilon-okt3-muromonab>

Anti-CD3D monoclonal antibody, clone PLU4 (DCABH-10124). Product Information. CD Creative Diagnostics. Publication date unknown. 2 pages. cited by applicant

Anti-TCR [BMA031]. Absolute Antibody. Website. Copyright 2021. Retrieved Dec. 26, 2021 at URL: <https://absoluteantibody.com/product/anti-tcr-bma031/>

Anwer et al., Donor origin CAR T cells: graft versus malignancy effect without GVHD, a systematic review, *Immunotherapy*, Jan. 2017;9(2):123-130. cited by applicant

Appelbaum. Haematopoietic cell transplantation as immunotherapy. *Nature*, vol. 411, pp. 385-389 (2001). cited by applicant

Arafat et al. Antineoplastic effect of anti-erbB-2 intrabody is not correlated with scFv affinity for its target. *Cancer Gene Therapy*, vol. 7, No. 9, 2000: pp. 125-130. cited by applicant

Arase et al. Recognition of virus infected cells by NK cells (w/ English abstract). Department of Immunochemistry, Research Institute for Microbial Diseases, Osaka University. cited by applicant

Austyn et al. T cell activation by anti-CD3 antibodies: function of Fc receptors on B cell blasts, but not resting B cells, and CD18 on the responding T cells. *European Journal of Immunology*, 1982;12(1):1-6. cited by applicant

Böldicke et al. Functional inhibition of transitory proteins by intrabody-mediated retention in the endoplasmic reticulum. *Methods* 56 (2012) 338-350. Available online Aug. 17, 2011. cited by applicant

Bleakley et al., Molecules and mechanisms of the graft-versus-leukaemia effect, *Nature Reviews, Cancer*, May 2004;4(5):371-80. cited by applicant

Boettcher et al., Choosing the Right Tool for the Job: RNAi, TALEN, or CRISPR, *Mol Cell*. May 21, 2015;58:575-85. cited by applicant

Brentjens et al. CD19-targeted T cells rapidly induce molecular remissions in adults with chemotherapy-refractory acute lymphoblastic leukemia. *Sci Transl Med* 3(10):245ra65 (2011). cited by applicant

Brentjens et al. Eradication of systemic B-cell tumors by genetically targeted human T lymphocytes co-stimulated by CD80 and interleukin-15. *Nat Med*. Mar. 2013;19(3):355-360. cited by applicant

Brentjens et al. Safety and persistence of adoptively transferred autologous CD19-targeted T cells in patients with relapsed or chemotherapy refractory B-cell acute lymphoblastic leukemia. *Nat Med*. Aug. 17, 2011. cited by applicant

Brudno et al. Allogeneic T cells that express an anti-CD19 chimeric antigen receptor induce remissions of B-cell malignancies that progress after allogeneic host-versus-host disease. *J Clin Oncol* 34(10):1112-1121 (2016). cited by applicant

Burns et al. Two monoclonal anti-human T lymphocyte antibodies have similar biologic effects and recognize the same cell surface antigen. *J Immunol* 1982;129(1):1-6. cited by applicant

Campana et al. 4-1BB chimeric antigen receptors. *Cancer J. Mar.-Apr. 2014;20(2):134-40*. cited by applicant

Ceuppens et al. Failure of OKT3 monoclonal antibody to induce lymphocyte mitogenesis: a familial defect in monocyte helper function. *J Immunol*, vol. 134, pp. 100-104 (1985). cited by applicant

Chang et al. A chimeric receptor with NKG2D specificity enhances natural killer cell activation and killing of tumor cells. *Cancer Res* 73(6):1777-86 (Mar. 15, 2013). cited by applicant

Chen et al., Donor-derived CD19-targeted T cell infusion induces minimal residual disease-negative remission in relapsed B-cell acute lymphoblastic leukemia after haploidentical haematopoietic stem cell transplantation. *British Journal of Haematology*, 2017, 179, 598-605. Epub Oct. 26, 2017. cited by applicant

Clevers et al., The T cell receptor/CD3 complex: a dynamic protein ensemble, *Annual Review of Immunology*, 1988;6:629-62. cited by applicant

Cooley et al. Donor selection for natural killer cell receptor genes leads to superior survival after unrelated transplantation for acute myelogenous leukemia. *Blood*. Feb. 15, 2012. cited by applicant

Cooper, et al. T-cell clones can be rendered specific for CD19: toward the selective augmentation of the graft-versus-B-lineage leukemia effect. *Blood*. Feb. 15, 2012. cited by applicant

Co-pending U.S. Appl. No. 17/862,721, inventors Campana; Dario et al., filed Jul. 12, 2022. cited by applicant

Co-pending U.S. Appl. No. 17/862,797, inventors Campana; Dario et al., filed Jul. 12, 2022. cited by applicant

Dai et al., Tolerance and efficacy of autologous or donor-derived T cells expression CD19 chimeric antigen receptors in adult B-ALL with extramedullary leukemia. *Journal of Immunotherapy*. cited by applicant

Davila. Efficacy and toxicity management of 19-28z CAR T cell therapy in B cell acute lymphoblastic leukemia. *Sci Transl Med* 6(224):224ra25 (2014). cited by applicant

EP16746922.0 Extended European Search Report dated Sep. 21, 2018. cited by applicant

EP18844536.5 Extended European Search Report dated Mar. 18, 2021. cited by applicant

Eshhar et al. Specific activation and targeting of cytotoxic lymphocytes through chimeric single chains consisting of antibody-binding domains and the gamma chain of the T cell receptor. *Proc Natl Acad Sci U S A*, vol. 90, pp. 720-724 (Jan. 1993). cited by applicant

Eyquem et al., Targeting a CAR to the TRAC locus with CRISPR/Cas9 enhances tumour rejection, *Nature*. Mar. 2, 2017; 543(7643): 113-117. cited by applicant

Gale. Cellular and Molecular Basis of Cancer. Merck Manual Professional Version. Last full review/revision Nov. 2020. Content last modified Nov. 2020. Retrieved Aug. 22, 2022 at URL: <https://www.merckmanuals.com/professional/hematology-and-oncology/overview-of-cancer/cellular-and-molecular-basis-of-cancer#8> 8 pages. cited by applicant

Gan et al. Molecular Mechanisms and Potential Therapeutic Reversal of Pancreatic Cancer-Induced Immune Evasion. *Cancers* 2020, 12, 1872. Published Jul. 2020. cited by applicant

Gao et al. Retention mechanisms for ER and Golgi membrane proteins. *Trends in Plant Science*, vol. 19, Issue 8, pp. 508-515 (Aug. 2014). cited by applicant

Geiger et al. Integrated src kinase and costimulatory activity enhances signal transduction through single-chain chimeric receptors in T lymphocytes. *Blood* 98(12):3653-3661 (2001). cited by applicant

Geiger et al. The TCR zeta-chain immunoreceptor tyrosine-based activation motifs are sufficient for the activation and differentiation of primary T lymphocytes. *J Immunol* 164(1):1-10 (2000). cited by applicant

Ghosh et al., Donor CD19 CAR T cells exert potent graft-versus-lymphoma activity with diminished graft-versus-host activity, *Nat Med*. Feb. 2017 ; 23(2): 243-250. cited by applicant

Giebel et al. Survival advantage with KIR ligand incompatibility in hematopoietic stem cell transplantation from unrelated donors. *Blood* (2003) 102 (3): 814-820. cited by applicant

Grimshaw. Developing a universal T cell for use in adoptive immunotherapy (thesis), University College London (2015). Retrieved Aug. 22, 2022 at URL: <http://discovery.ucl.ac.uk/1470207/1/Grimshaw%20Ben%20Thesis.pdf>. 267 pages. cited by applicant

Grimshaw et al. Creating a Null T Cell for Adoptive Immunotherapy. Poster. Mar. 9, 2012. One page. cited by applicant

Grovender et al. Single-chain antibody fragment-based adsorbent for the extracorporeal removal of β_2 -microglobulin. *Kidney International*, vol. 65 (2004), pp. 105-112. cited by applicant

Grupp et al. Chimeric Antigen Receptor-Modified T Cells for Acute Lymphoid Leukemia. *N Engl J Med* 368; 16, pp. 1509-1508 (Apr. 18, 2013). With correct citation. cited by applicant

Haegert et al. Co-expression of surface immunoglobulin and T3 on hairy cells. *Scand J Haematol* 1986;37:196-202. cited by applicant

Haynes et al. Rejection of syngeneic colon carcinoma by CTLs expressing single-chain antibody receptors codelivering CD28 costimulation. *J Immunol* 2002;168(1):100-106. cited by applicant

Haynes et al. Single-chain antigen recognition receptors that costimulate potent rejection of established experimental tumors. *Blood*. Nov. 1, 2002;100(9):3151-3157. cited by applicant

Hegde, M. et al., Supplementary Material for "Combinational targeting offsets antigen escape and enhances effector functions of adoptively transferred T cells" (2015). cited by applicant

2101. Retrieved Jan. 6, 2022 at URL: <https://ars.elsa-cdn.com/content/image/1-s2.0-S1525001616309315-mmc1.pdf>. 9 pages. cited by applicant

Hexham et al. Influence of relative binding affinity on efficacy in a panel of anti-CD3 scFv immunotoxins. *Molecular Immunology* 38 (2001) 397-408. cited by applicant

Holliger et al. "Diabodies": small bivalent and bispecific antibody fragments. *Proc Natl Acad Sci USA*, vol. 90, pp. 6444-6448 (Jul. 1993). cited by applicant

Hombach et al. Tumor-Specific T Cell Activation by Recombinant Immunoreceptors: CD3Z Signaling and CD28 Costimulation Are Simultaneously Required Combined CD28/CD3ζ Signaling Receptor Molecule. *J Immunol* 2001; 167:6123-6131. cited by applicant

Imai et al. Chimeric receptors with 4-1BB signaling capacity provoke potent cytotoxicity against acute lymphoblastic leukemia. *Leukemia* (2004) 18, 676-684

Imai et al. Genetic modification of primary natural killer cells overcomes inhibitory signals and induces specific killing of leukemic cells. *Blood* (2005) 106 (3): 1063-1068

Imamura. M. et al., Supplementary Material for "Autonomous growth and increased cytotoxicity of natural killer cells expressing membrane-bound interleukin-2". Retrieved Jan. 6, 2022 at URL: https://ash.silverchair-cdn.com/ash/content_public/journal/blood/124/7/10.1182_blood-2014-02-556837/4/blood-2014-02-556837-1.pdf?Expires=1644340511&Signature=uyyKt0KS8WtXpVUILTgok2RyynmzBJgE2vCNIDD4xdwCv13vsg0goCLOpQLU^KVPtTvlHtmCLEX2MhA7mcxzj-cM-MgQfrETegs4oK6vjiveJODaZP6TfW1gGK^5JUAn5LesZfPv9W28NmBfMoAOMVeX4Pz54V^9dWaBcCfXCR7vOrx1N8cpbxm1AumSziwKqxNCy79c-giQZhSxfLm2ctwuCbOMj8RIHCM4cO5a2zIKMAM6dut-dafGbQaRQ_&Key-Pair-Id=APKAIE5G5CRDK6RD3PGA. 11 pages. cited by applicant

Jackson et al., Identification of a consensus motif for retention of transmembrane proteins in the endoplasmic reticulum, *Embo J.* Oct. 1990; 9(10): 3153-3162

Joshi et al., T cells with a highly charged proteasomal retargeting sequence increases soluble cytoplasmic expression and efficacy of diverse anti-synuclein intrabodies. *PLoS One* 7(12): e49607. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1371/journal.pone.0049607>. 12 pages. cited by applicant

online: Aug. 28, 2012. cited by applicant

Kalos, et al., "T Cells with Chimeric Antigen Receptors Have Potent Antitumor Effects and Can Establish Memory in Patients with Advanced Leukemia", *Sci Transl Med* 4(161): 257ra121. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1126/scitranslmed.1257ra121>. cited by applicant

Kamiya et al. Blocking expression of inhibitory receptor NKG2A overcomes tumor resistance to NK cells. *J Clin Invest.* 2019; 129(5):2094-2106. cited by applicant

Kloss, C.C. et al., Supplementary Text and Figures for "Combinatorial antigen recognition with balanced signaling promotes selective tumor eradication by engineered T cells". Retrieved Jan. 6, 2022 at URL: https://static-content.springer.com/esm/art%3A10.1038%2Fnb.2459/MediaObjects/41587_2013_BFnb2459_MOESM2_1.pdf. 11 pages. cited by applicant

Kloss. Combinatorial antigen recognition with balanced signaling promotes selective tumor eradication by engineered T cells. *Nature Technology*, vol. 31, No. 12, pp. 1247-1252. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1038/natech.2013.12>. cited by applicant

Kochenderfer et al. B-cell depletion and remissions of malignancy along with cytokine-associated toxicity in a clinical trial of anti-CD19 chimeric-antigen-receptor-expressing T cells in patients with B-cell lymphoma. *N Engl J Med* 366(10): 857-867. Published online Dec. 8, 2011. cited by applicant

Kochenderfer et al., Chemotherapy-refractory diffuse large B-cell lymphoma and indolent B-cell malignancies can be effectively treated with autologous T cells expressing anti-CD19 chimeric antigen receptor. *N Engl J Med* 366(10): 857-867. Published online Aug. 25, 2014. cited by applicant

Kochenderfer et al., Donor-derived CD19-targeted T cells cause regression of malignancy persisting after allogeneic hematopoietic stem cell transplantation, *Blood* 120(12): 3105-3113. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1182/blood-2012-05-428217>. cited by applicant

Kolb et al. Graft-versus-leukemia effect of donor lymphocyte transfusions in marrow grafted patients. *Blood*, vol. 86, No. 5 (Sep. 1), 1995: pp. 2041-2050. cited by applicant

Kotteas et al. Immunotherapy for pancreatic cancer. *J Cancer Res Clin Oncol* (2016) 142:1795-1805. Published online Feb. 3, 2016. cited by applicant

Lee, et al. Current concepts in the diagnosis and management of cytokine release syndrome. *Blood*. Jul. 10, 2014;124(2):188-95. doi: 10.1182/blood-2014-05-1234567. cited by applicant

Lee et al., T cells expressing CD 19 chimeric antigen receptors for acute lymphoblastic leukaemia in children and young adults: a phase 1 dose-escalation trial. *Lancet* 386(10001): 231-236. Retrieved Jan. 6, 2022 at URL: [https://doi.org/10.1016/S0140-6736\(15\)00454-2](https://doi.org/10.1016/S0140-6736(15)00454-2). cited by applicant

2014. cited by applicant

Lo et al. Harnessing the tumour-derived cytokine, CSF-1, to co-stimulate T-cell growth and activation. *Molecular Immunology* 45 (2008) 1276-1287. Available at: <https://doi.org/10.1016/j.molimm.2008.05.008>. cited by applicant

Lorentzen et al. CD19-Chimeric Antigen Receptor T Cells for Treatment of Chronic Lymphocytic Leukaemia and Acute Lymphoblastic Leukaemia. *Scandinavian Journal of Immunology* 77(1): 1-10. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1111/j.1365-3113.2012.04811.x>. cited by applicant

applicant

Maciocia et al. A simple protein-based method for generation of 'off the shelf' allogeneic chimeric antigen receptor T-cells. *Database Embase [Online]* 1-15, EMBASE 2012/2784541. Database Accession No. EMB-623339718. cited by applicant

Maciocia et al. A simple protein-based method for generation of 'off the shelf' allogeneic chimeric antigen receptor T-cells. *Molecular Therapy*, vol. 26, No. 5, pp. 1515-1524. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1038/mt.2013.12>. cited by applicant

16, 2018 to May 19, 2018 Chicago, IL-297 Conf. ISSN: 1525-0024. cited by applicant

MacLeod et al., Integration of a CD19 CAR into the TCR Alpha Chain Locus Streamlines Production of Allogeneic Gene-Edited CAR T Cells, *Molecular Therapy* 26(12): 2315-2324. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1016/j.molimm.2018.10.008>. cited by applicant

Maher et al. Human T-lymphocyte cytotoxicity and proliferation directed by a single chimeric TCRzeta /CD28 receptor. *Nat Biotech* 20(1):70-75 (2002). cited by applicant

Marasco et al. Design, intracellular expression, and activity of a human anti-human immunodeficiency virus type 1 gp120 single-chain antibody. *PNAS USA* 99(12): 15515-15520. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1073/pnas.120155199>. cited by applicant

Maude, et al., Chimeric Antigen Receptor T Cells for Sustained Remissions in Leukemia. *N. Engl J. Med* (2014) 371(16) 1507-1517. With correction published in *N. Engl J. Med* 372(12): 1157-1158. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1056/NEJMoa1300419>. cited by applicant

Miller et al. Successful adoptive transfer and in vivo expansion of human haploidentical NK cells in patients with cancer. *Blood* 105:3051-3057 (2005). cited by applicant

Miller. Therapeutic applications: natural killer cells in the clinic. *Hematology Am Soc Hematol Educ Program* (2013) 2013 (1): 247-253. cited by applicant

Munro et al., A C-terminal signal prevents secretion of luminal ER proteins, *Cell*. Mar. 13, 1987;48:899-907. Retrieved Apr. 7, 2022 at URL: <https://bio.davidsonlab.org/papers/munro1987.pdf>. cited by applicant

Kochenderfer et al., Eradication of B-lineage cells and regression of lymphoma in a patient treated with autologous T cells genetically engineered to recognize CD19. *N Engl J Med* 366(10): 883-892. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1056/NEJMoa1300419>. cited by applicant

applicant

Needleman et al. A general method applicable to the search for similarities in the amino acid sequence of two proteins. *Journal of molecular biology* 48(3):443-453. Retrieved Jan. 6, 2022 at URL: [https://doi.org/10.1016/0022-2730\(73\)90133-9](https://doi.org/10.1016/0022-2730(73)90133-9). cited by applicant

Neelapu et al., Axicabtagene Ciloleucel CAR T-Cell Therapy in Refractory Large B-Cell Lymphoma, *N Engl J Med* 377:26 pp. 2531-2544 (Dec. 28, 2017). cited by applicant

Pardoll. The blockade of immune checkpoints in cancer immunotherapy. *Nat Rev Cancer*. 12(4):252-264 (2012). cited by applicant

Park et al. Are all chimeric antigen receptors created equal? *J Clin Oncol.* Feb. 20, 2015;33(6):651-3. Epub Jan. 20, 2015. cited by applicant

Park et al., CD19-targeted CAR T-cell therapeutics for hematologic malignancies: interpreting clinical outcomes to date, *Blood*. 2016;127(26):3312-3320. cited by applicant

PCT/SG2016/050063 International Search Report and Written Opinion dated May 9, 2016. cited by applicant

PCT/US2018/046137 International Search Report and Written Opinion dated Oct. 29, 2018. cited by applicant

Pearson and Lipman, Improved tools for biological sequence comparison, *Proc. Nat. Acad Sci USA.*, 85:2444-2448, (1988). cited by applicant

Peipp et al. A Recombinant CD7-specific Single-Chain Immunotoxin Is a Potent Inducer of Apoptosis in Acute Leukemic T Cells. *Cancer Research* 62, pp. 2833-2839. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1158/0008-5472.CCR-01-0585>. cited by applicant

Poirot et al. Multiplex Genome-Edited T-cell Manufacturing Platform for Off-the-Shelf Adoptive T-cell Immunotherapies. *Cancer Research* 75(18):3853-3864. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1158/0008-5472.CCR-15-0885>. cited by applicant

Porter et al., Chimeric antigen receptor-modified T cells in chronic lymphoid leukemia. *N Engl J Med* 365:725-733 (2011). With correction published in *N. Engl J. Med* 366(10): 1099-1100. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1056/NEJMoa1007538>. cited by applicant

Porter et al. Induction of Graft-versus-Host Disease as Immunotherapy for Relapsed Chronic Myeloid Leukemia. *N Engl J Med* 1994; 330:100-106. cited by applicant

Qasim et al., Molecular remission of infant B-ALL after infusion of universal TALEN gene-edited CART cells. *Sci. Transl. Med.* 9, eaaj2013 (2017). With Erratum. Retrieved Jan. 6, 2022 at URL: <https://www.science.org/doi/10.1126/scitranslmed.aam9292>. 3 pages. cited by applicant

Reshef et al. Blockade of lymphocyte chemotaxis in visceral graft-versus-host disease. *N Engl J Med* 367;3 pp. 135-145 (Jul. 12, 2012). cited by applicant

Rosenberg et al. Adoptive cell transfer as personalized immunotherapy for human cancer. *Science* 348(6230):62-68 (2015). cited by applicant

Rowley, J. et al., Supplementary Information for "Expression of IL-15RA or an IL-15/IL-15RA fusion on CD8+ T cells modifies adoptively transferred T-cell killing of leukemia cells". Retrieved Jan. 6, 2022 at URL: https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1002%2Ffej.200838594&file=ej_200838594.pdf. cited by applicant

pp. 491-506. Retrieved Jan. 6, 2022 at URL: https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1002%2Ffej.200838594&file=ej_200838594.pdf. cited by applicant

Rubnitz et al. NKAML: A Pilot Study to Determine the Safety and Feasibility of Haploidentical Natural Killer Cell Transplantation in Childhood Acute Myeloid Leukemia. *Transplantation* 97(1): 101-108. Retrieved Jan. 6, 2022 at URL: <https://doi.org/10.1097/TP.0000000000000101>. cited by applicant

by applicant

Rudikoff, et al. Single amino acid substitution altering antigen-binding specificity. *Proc Natl Acad Sci U S A.* Mar. 1982;79(6):1979-83. cited by applicant

Ruggeri et al. Effectiveness of donor natural killer cell alloreactivity in mismatched hematopoietic transplants. *Science* 295(5562):2097-2100 (2002). cited by applicant

Sadelain et al. The Basic Principles of Chimeric Antigen Receptor Design. *Cancer Discov.* Apr. 2013 ; 3(4): 388-398. cited by applicant

Sadelain et al. Therapeutic T cell engineering. *Nature* 545:423-431 (2017). cited by applicant

Sato et al. Single domain intrabodies against WASP inhibit TCR-induced immune responses in transgenic mice T cells. *Sci Rep.* Oct. 21, 2013;3:3003. 10 pages. cited by applicant

Schumann, et al. Generation of knock-in primary human T cells using Cas9 ribonucleoproteins. *Proc Natl Acad Sci U S A.* Aug. 18, 2015;112(33):10437-42. cited by applicant

applicant

Shuster et al., Chimeric Antigen Receptor T Cells in Refractory B-Cell Lymphomas, Dec. 28, 2017, the New England Journal of Medicine, 2017; 377; pp. 23-31.

Schwartz. T cell anergy. *Annu Rev Immunol.* 2003;21:305-34. First published online as a Review in Advance on Dec. 5, 2002. cited by applicant

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-16.rag. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=bc544287-1664-44c7-9aff-7ec7ba8cb7ea&itemName=20220829_091720_... 14 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-16.rai. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=131a34de-dcc5-448f-b75f-7ea24f699275&itemName=20220829_091720_... 8 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-16.rapm. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=e6f49583-c02c-4eee-a87c-e0ece396955c&itemName=20220829_091720_... 10 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-17.rag. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=cbb41ae5-1511-498f-ac7c-0b33abd2492f&itemName=20220829_091720_... 15 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-17.rai. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=73e892a9-d59f-4aff-8df1-883c983c2966&itemName=20220829_091720_u... 8 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-17.rapm. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=9851a8bc-7a77-4e69-985e-903a1f2622ae&itemName=20220829_09172_... 10 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-17.rpr. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=e04b2be2-c5c8-4cba-b806-cd02a1efe868&itemName=20220829_091720_... 7 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-20.rai. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=d17ad86f-3b93-47e9-a94e-a9258afaa3e3&itemName=20220829_091720_... 9 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-20.rapm. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=5747899e-d2f6-4407-8197-6ae6e43ac551&itemName=20220829_09172_... 11 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-20.rpr. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=8db4eb9f-566a-448b-8a1b-9244e2b8f0ee&itemName=20220829_091720_... 7 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-21.rag. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=69e4190a-317d-4f26-9846-c30ab0601669&itemName=20220829_09172_... 14 pages.

Score Search Results Detail for U.S. Appl. No. 17/862,721 and Search Results Aug. 29, 2022_091720_us-17-862-721-21.rai. Search run on Aug. 29, 2022. http://score.uspto.gov/ScoreAccessWeb/getItemDetail?appId=17862721&docId=a9c72e86-3699-4e4c-8110-8367d0fe6cf4&itemName=20220829_091720_... 8 pages.

Sharma et al., "Novel cancer immunotherapy agents with survival benefit: recent successes and next steps," *Nat Rev Cancer*, 11(11):805-812 (2011). cited by applicant

Shikano et al., Membrane receptor trafficking: Evidence of proximal and distal zones conferred by two independent endoplasmic reticulum localization signals," *J Biol Chem*, 281(12):7445-7454 (2006). cited by applicant

Shimasaki et al. A clinically adaptable method to enhance the cytotoxicity of natural killer cells against B-cell malignancies. *Cytotherapy*, 2012; 14: 830-840. cited by applicant

Shimasaki et al., Natural killer cell reprogramming with chimeric immune receptors, *Methods Mol Biol.* 2013;969:203-20. cited by applicant

Slavin et al. Allogeneic cell therapy with donor peripheral blood cells and recombinant human interleukin-2 to treat leukemia relapse after allogeneic bone marrow transplantation. *Transplantation*, 2003; 76(12):1985-1991. cited by applicant

Smith et al. Comparison of Biosequences. *Advances in Applied Mathematics.* 2:482-489 (1981). cited by applicant

Smith et al. T cell activation by anti-T3 antibodies: Comparison of IgG1 and IgG2b switch variants and direct evidence for accessory function of macrophage activation 1. *J Immunol*, 1981; 126(1):105-112. cited by applicant

Sommermeier et al., Fully human CD19-specific chimeric antigen receptors for T-cell therapy. *Leukemia*. Oct. 2017 ; 31(10): 2191-2199. cited by applicant

Strebe et al. Functional knockdown of VCAM-1 at the posttranslational level with ER retained antibodies. *Journal of Immunological Methods* 341 (2009) 30-40. cited by applicant

Su et al., CRISPR-Cas9 mediated efficient PD-I disruption on human primary T cells from cancer patients, *Sci Rep* 6, 20070, Published: Jan. 28, 2016. 14 pages. cited by applicant

Till et al. CD20-specific adoptive immunotherapy for lymphoma using a chimeric antigen receptor with both CD28 and 4-1BB domains: pilot clinical trial results. *Journal of Immunotherapy*, 2012; 35(1): 1-10. cited by applicant

Todorovska et al. Design and application of diabodies, triabodies and tetrabodies for cancer targeting. *Journal of Immunological Methods* 248 (2001) 47-66. cited by applicant

Topp et al. Targeted therapy with the T-cell-engaging antibody blinatumomab of chemotherapy-refractory minimal residual disease in B-lineage acute lymphoblastic leukemia: results of a phase II trial. *J Clin Oncol.* Jun. 20, 2011;29(18):2493-8. cited by applicant

Torikai et al. A foundation for universal T-cell based immunotherapy: T cells engineered to express a CD19-specific chimeric-antigen-receptor and eliminate B-cell malignancies. *Journal of Immunotherapy*, 2012; 35(1): 1-10. cited by applicant

Turtle et al. CD19 CAR-T cells of defined CD4+:CD8+ composition in adult B cell ALL patients. *J Clin Invest* 126(6):2123-2138 (2016). cited by applicant

U.S. Appl. No. 15/548,577 Notice of Allowance dated May 4, 2020. cited by applicant

U.S. Appl. No. 15/548,577 Office Action dated Jan. 14, 2019. cited by applicant

U.S. Appl. No. 15/548,577 Office Action dated Jul. 20, 2018. cited by applicant

U.S. Appl. No. 15/548,577 Office Action dated Sep. 17, 2019. cited by applicant

U.S. Appl. No. 16/100,117 Office Action dated Aug. 12, 2022. cited by applicant

U.S. Appl. No. 16/100,117 Office Action dated Dec. 9, 2021. cited by applicant

U.S. Appl. No. 16/100,117 Office Action dated May 3, 2021. cited by applicant

U.S. Appl. No. 16/100,120 Office Action dated Aug. 2, 2019. cited by applicant

U.S. Appl. No. 16/100,120 Office Action dated Feb. 24, 2020. cited by applicant

U.S. Appl. No. 16/100,120 Office Action dated Jan. 28, 2019. cited by applicant

U.S. Appl. No. 16/100,120 Office Action dated Oct. 6, 2020. cited by applicant

U.S. Appl. No. 17/862,721 Office Action dated Sep. 7, 2022. cited by applicant

Van Wauwe et al. Human T lymphocyte activation by monoclonal antibodies; OKT3, but not UCHT1, triggers mitogenesis via an interleukin 2-dependent mechanism. *J Immunol*, 1981; 126(1):105-112. cited by applicant

Verneris et al. Natural Killer Cell Consolidation for Acute Myelogenous Leukemia: A Cell Therapy Ready for Prime Time? *J Clin Oncol.* Feb. 20, 2010; 28(6):1055-1062. cited by applicant

Verwilghen et al. Differences in the stimulating capacity of immobilized anti-CD3 monoclonal antibodies: variable dependence on interleukin-1 as a helper signal. *J Immunol*, 1981; 126(1):105-112. cited by applicant

Vivier, Eric et al., Innate or adaptive immunity? The example of natural killer cells, *Science* (New York, N.Y.) vol. 331,6013 (2011): 44-9. doi:10.1126/science.1199313. cited by applicant

Wang et al. Expression and characterization of recombinant soluble monkey CD3 molecules: mapping the FN18 polymorphic epitope. *Molecular Immunology*, 2003; 40(12):1055-1062. cited by applicant

Weetall et al. T-cell depletion and graft survival induced by anti-human CD3 immunotoxins in human CD3ε transgenic mice. *Transplantation*, vol. 73, 1658-1664 (2002). cited by applicant

Wheeler et al. Intrabody and intrakine strategies for molecular therapy. *Molecular Therapy*, vol. 8, No. 3, pp. 355-366, Sep. 2003. cited by applicant

Wunderlich et al. OKT3 prevents xenogeneic GVHD and allows reliable xenograft initiation from unfractionated human hematopoietic tissues. *Blood*. 2014; 124(12):1985-1991. cited by applicant

Yang et al., Challenges and opportunities of allogeneic donor-derived CAR T cells, *Current Opinion in Hematology*, Nov. 2015; 22 (6); pp. 509-515. cited by applicant

Zang et al. The B7 family and cancer therapy: costimulation and coinhibition. *Clin Cancer Res* 13(18) pp. 5271-5279 (Sep. 15, 2007). cited by applicant

Zhan et al. Modification of ricin A chain, by addition of endoplasmic reticulum (KDEL) or Golgi (YQRL) retention sequences, enhances its cytotoxicity and targets it to the Golgi apparatus. *J Biol Chem*, 273(1):55-60 (1998). cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_143513_us-15-548-577-37.rag. Search run on Jan. 7, 2019. Copy <http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b67...> 4 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_143514_us-15-548-577-36.rapbm. Search run on Jan. 7, 2019. Copy <http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b67...> 4 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_143514_us-15-548-577-37.rag. Search run on Jan. 7, 2019. Copy r
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 21 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_143514_us-15-548-577-36.rapbm. Search run on Jan. 7, 2019. Co
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 13 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_143514_us-15-548-577-37.rai. Search run on Jan. 7, 2019. Copy r
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 11 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_145543_us-15-548-577-32.rag. Search run on Jan. 7, 2019. Copy
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 23 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_145544_us-15-548-577-32.rai. Search run on Jan. 7, 2019. Copy r
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 4 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_145544_us-15-548-577-32.rapbm. Search run on Jan. 7, 2019. Co
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 6 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_145544_us-15-548-577-33.rai. Search run on Jan. 7, 2019. Copy r
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 4 pages. cited by applicant

Score Search Results Details for U.S. Appl. No. 15/548,577 and Search Result Jan. 7, 2019_145544_us-15-548-577-33.rapbm. Search run on Jan. 7, 2019. Co
<http://score.uspto.gov/ScoreAccessWeb/getItem.htm?AppId=15548577&seqId=09323b678> . . . 12 pages. cited by applicant

XP-002784541 “A simple protein-based method for generation of ‘off the shelf’ allogeneic chimeric antigen receptor T-cells”, Journal Conference Abstract, M
by applicant

Arase, Hisashi, et al. “Recognition of virus infected cells by NK cells”, Department of Immunochemistry, Research Institute for Microbial Diseases, Osaka Un
Böldicke, Thomas, Blocking translocation of cell surface molecules from the ER to the cell surface by intracellular antibodies targeted to the ER, J. Cell. Mol.
Clift, Dean, et al “A Method for the Acute and Rapid Degradation of Endogenous Proteins”, Elsevier Inc., Cell 172, Dec. 14, 2017, pp. 1692-1706. cited by ap
Dotti, et al., 2014 Design and development of therapies using chimeric antigen receptor-expressing T cells Immunological Rev, pp. 107-126. cited by applicant

Grimshaw, B.D. et al., BGST Abstract Mar. 9, 2012, abstract P023, “Creating a ‘null’ T cell for use in adoptive immunotherapy”, British Society for Gene and
22 pages. cited by applicant

Hegde, M. et al., “Combinational targeting offsets antigen escape and enhances effector functions of adoptively transferred T cells in glioblastoma”, Molecula
Kamiya, et al., A novel method to generate T-cell receptor-deficient antigen receptor T cells, Blood Advances, vol. 2, No. 5, Mar. 13, 2018, pp. 517-528. cited
Kloss, C.C. et al., “Combinatorial antigen recognition with balanced signaling promotes selective tumor eradication by engineered T cells”, Nature Biotechnol
Kudo, K. et al., “T lymphocytes expressing a CD16 signaling receptor exert antibody dependent cancer cell killing”. Cancer Research, 2014, vol. 74, pp. 93-1
Lantis, E. et al., “Chimeric antigen receptor T Cells with dissociated signaling domains exhibit focused antitumor activity with reduced potential for toxicity in
cited by applicant

Lo, A.S.Y. et al., “Harnessing the tumour-derived cytokine, CSF-1, to co-stimulate Tcell growth and activation”, Molecular Immunology, 2008, vol. 45, pp. 12
Imamura. M. et al., “Autonomous growth and increased cytotoxicity of natural killer cells expressing membrane-bound interleukin-15”, Blood, 2014, vol. 124
Marschall, Andrea LJ, et al “Specific in vivo knockdown of protein function by intrabodies”, Taylor & Francis Group, LLC, Nov./Dec. 2015, vol. 7, Issue 6, p
Milone, Michael C., et al. “Chimeric Receptors Containing CD137 Signal Transduction Domains Mediate Enhanced Survival of T Cells and Increased Antileu
1453-1464. cited by applicant

Png, et al., “Blockade of CD7 expression in T cells for effective chimeric antigen receptor targeting of T-cell malignancies”, Blood Advances, vol. 1, No. 25, M
Rossig, C. et al., “Genetic modification of T lymphocytes for adoptive immunotherapy”, Molecular Therapy, 2004, vol. 10, pp. 5-18. cited by applicant

Rowley, J. et al., “Expression of IL-15RA or an IL-15/IL-15RA fusion on CD8+ T cells modifies adoptively transferred T-cell function in cis”, European Jour
Sanz, L. et al., “Antibodies and gene therapy: teaching old ‘magic bullets’ new tricks”, Trends in Immunology, 2004, vol. 25, pp. 85-91. cited by applicant

Zhou, P. et al., “Cells transfected with a non-neutralizing antibody gene are resistant to HIV infection: targeting the endoplasmic reticulum and *trans-Golgi* ne
1496. cited by applicant

Grimshaw, Benjamin David “Developing a universal T cell for use in adoptive immunotherapy”, XP055493578, Jan. 1, 2015, pp. 190-197, retrieved from inte
URL:<http://discovery.ucl.ac.uk/1470207/1/Grimshaw%20Ben%20Thesis.pdf>. cited by applicant

Brown et al. Novel Treatments for Chronic Lymphocytic Leukemia and Moving Forward. American Society of Clinical Oncology Educational Book, pp. e317
https://ascopubs.org/doi/pdfdirect/10.14694/EdBook_AM.2014.34.e317. cited by applicant

Caruana et al. From Monoclonal Antibodies to Chimeric Antigen Receptors for the Treatment of Human Malignancies. Semin Oncol. Oct. 2014 ; 41(5): 661-6
Certified PCT/US2016/019953 priority document U.S. Appl. No. 62/121,842, Ma, Yupo et al., inventors, filed Feb. 27, 2015. cited by applicant

Feng et al. Treatment of Aggressive T Cell Lymphoblastic Lymphoma/leukemia Using Anti-CD5 CAR T Cells. Stem Cell Reviews and Reports (2021) 17:652
Fujiwara, et al. Adoptive Immunotherapy for Hematological Malignancies Using T Cells Gene-Modified to Express Tumor Antigen-Specific Receptors. Pharm
applicant

Glienke et al. Advantages and applications of CAR-expressing natural killer cells. Frontiers in Pharmacology, vol. 6, Article 21 (Feb. 12, 2015). 7 pages. cited
Guo et al. Efficiency and side effects of anti-CD38 CAR T cells in an adult patient with relapsed B-ALL after failure of bi-specific CD19/CD22 CAR T cell tr
applicant

Hishima et al. CD5 Expression in Thymic Carcinoma. American Journal of Pathology, vol. 145, No. 2, pp. 268-275 (Aug. 1994). cited by applicant

Imboden et al. Stimulation of CD5 enhances signal transduction by the T cell antigen receptor. J Clin Invest. 1990. 85:130-134. cited by applicant

Jena et al., Redirecting T-cell specificity by introducing a tumor-specific chimeric antigen receptor. Blood. 116:1035-1044 (2010). cited by applicant

Kapp et al. Chapter 1: Post-Targeting Functions of Signal Peptides. In Protein Transport into the Endoplasmic Reticulum, Zimmerman, ed., 2009, Landes Bios
<https://www.ncbi.nlm.nih.gov/books/NBK6322/>. cited by applicant

Lemaistre et al. Phase I Trial of H65-RTA Immunoconjugate in Patients With Cutaneous T-cell Lymphoma. Blood, vol. 78, No. 5 (Sep. 1), 1991: pp. 1173-118
Lewis et al. The immunophenotype of pre-TALL/LBL revisited. Experimental and Molecular Pathology 81 (2006) 162-165. Available online Aug. 14, 2006. c
Li et al. Flow Cytometry in the Differential Diagnosis of Lymphocyte-Rich Thymoma From Precursor T-Cell Acute Lymphoblastic Leukemia/Lymphoblastic
applicant

Litzow et al. How I treat T-cell acute lymphoblastic leukemia in adults. Blood. 2015; 126(7):833-841. Retrieved at URL: <https://www.academia.edu/download>
cited by applicant

Liu et al. [NK cell surface receptors and their research progress—review]. Zhongguo shi yan xue ye xue za zhi. Aug. 2012;20(4):1034-1038. English abstract
<https://pubmed.ncbi.nlm.nih.gov/22931679/>. One page. cited by applicant

Morris et al. Antibody-based therapy of leukaemia. Expert Rev Mol Med. Sep. 30, 2009; 11:e29. cited by applicant

Pinz et al. Preclinical targeting of human T cell malignancies using CD4-specific chimeric antigen receptor (CAR)-engineered T cells. Leukemia, accepted art
https://www.researchgate.net/profile/Alexander-Jares/publication/283493430_Preclinical_targeting_of_human_T_cell_malignancies_using_CD4-specific_chi_engineered_T_cells/links/5643662808aef646e6c6a549/Preclinical-targeting-of-human-T-cell-malignancies-using-CD4-specific-chimeric-antigen-receptor-CA
Rezvani et al. The Application of Natural Killer Cell immunotherapy for the Treatment of Cancer. Frontiers in Immunology, vol. 6, Article 578 (Nov. 17, 2015)
Roitt et al. Roitt's Essential Immunology. 12th Edition, Wiley-Blackwell, Chapter 4 (2012). cited by applicant

Sommermeier, et al., Chimeric antigen receptor-modified T cells derived from defined CD8+ and CD4+ subsets confer superior antitumor reactivity in vivo. I
U.S. Appl. No. 17/862,721 Notice of Allowance dated Jun. 9, 2023. cited by applicant

Zhong-Fu et al. Clinical Applications of NK Cells in Tumor Immunotherapy. Chinese Journal of Biochemistry and Molecular Biology 27(12):1088-1093 (Dec

Arakawa et al. Cloning and Sequencing of the VH and YK Genes of an Anti-CD3 Monoclonal Antibody, and Construction of a Mouse/Human Chimeric Antic... applicant
Certificate of Disclosure by Benjamin D. Grimshaw signed Dec. 15, 2021 for: Grimshaw et al. Creating a Null T Cell for Adoptive Immunotherapy. Poster. M...
Certificate of Disclosure by David Lynch signed Dec. 16, 2021 for: Grimshaw et al. Creating a Null T Cell for Adoptive Immunotherapy. Poster. Mar. 9, 2012...
Certificate of Disclosure by Martin Pule signed Dec. 15, 2021 for: Grimshaw et al. Creating a Null T Cell for Adoptive Immunotherapy. Poster. Mar. 9, 2012...
Co-pending U.S. Appl. No. 18/295,226, inventors Campana; Dario et al., filed Apr. 3, 2023. cited by applicant
Declaration by Benjamin D. Grimshaw signed Sep. 17, 2022. 2 pages. cited by applicant
Nilsson et al. Retention and retrieval in the endoplasmic reticulum and the Golgi apparatus. *Current Opinion in Cell Biology* 1994, 6:517-521. cited by applica...
Dong, et al. *Modern Hematopoietic Stem Cell Transplant Therapeutics*. Military Health Press. 1st Edition (2001):175-176. With English machine translation. ...
GenBank Accession No. AAA83267. Version No. AAA83267.1. sFv antibody, partial [Mus musculus]. Record created Dec. 14, 1995. Page Range: 1-4. Retrie...
<https://www.ncbi.nlm.nih.gov/protein/AAA83267.1>. cited by applicant
GenBank Accession No. AAM12015. Version No. AAM12015.1. monoclonal anti-alpha-1,3-galactosyltransferase IgM heavy chain, partial [Mus musculus]. E...
2024 at URL: <https://www.ncbi.nlm.nih.gov/protein/AAM12015.1>. cited by applicant
GenBank Accession No. ABK63997. Version No. ABK63997.1. Immunoglobulin light chain variable region, partial [Mus musculus]. Record created Nov. 25, ...
<https://www.ncbi.nlm.nih.gov/protein/ABK63997>. cited by applicant
Chen, Kevin H., et al. Novel Anti-cd3 Chimeric Antigen Receptor Targeting of Aggressive T Cell Malignancies. *Oncotarget*, vol. 7, 56219-56232 (2016). cited...
Kipriyanov, Sergey M., et al. Two amino acid mutations in an anti-human CD3 single chain Fv antibody fragment that affect the yield on bacterial secretion b...
cited by applicant
U.S. Appl. No. 18/295,226 Office Action dated Aug. 2, 2024. cited by applicant
Ali Khan, Hyder, and Bulent Mutus. Protein disulfide isomerase a multifunctional protein with multiple physiological roles. *Frontiers in Chemistry* 2:70, 1-9 (...)
Janeway Jr, Charles A. et al. *T-cell receptor gene rearrangement. Immunobiology: The Immune System in Health and Disease*, 5th Edition. Garland Science :1...
Tragoolpua, Khajornsak. et al. Generation of functional scFv intrabody to abate the expression of CD147 surface molecule of 293A cells. *BMC biotechnology* ...
U.S. Appl. No. 17/862,797 Office Action dated Nov. 27, 2024. cited by applicant
EP22180254.9 Extended European Search Report dated Jan. 16, 2023. cited by applicant
U.S. Appl. No. 16/100,117 Notice of Allowance dated Feb. 17, 2023. cited by applicant
U.S. Appl. No. 16/100,117 Notice of Allowance dated Jan. 5, 2023. cited by applicant
U.S. Appl. No. 17/862,721 Notice of Allowance dated Feb. 15, 2023. cited by applicant
U.S. Appl. No. 17/862,721 Office Action dated Jan. 5, 2023. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/548,577, filed Aug. 3, 2017, now U.S. Pat. No. 10,765,699, which is a 371 U.S. National Phase Application of International Patent Cooperation Treaty Application PCT/SG2016/050063, filed Feb. 5, 2016, which claims benefit to U.S. Provisional Application No. 62/112,765, filed Feb. 6, 2015, and U.S. Provisional Application No. 62/130,970, filed Mar. 10, 2015, the disclosures of which are incorporated herein by reference in its entirety for all purposes.

REFERENCE TO A SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Aug. 2, 2017, is named 14445N SEQUENCE LISTING 8-2-17.TXT and is 32.43 kilobytes in size.

BACKGROUND OF THE INVENTION

(2) Immune cells can be potent and specific “living drugs”. Immune cells have the potential to target tumor cells while sparing normal tissues; several clinical observations indicate that they can have major anti-cancer activity. Thus, in patients receiving allogeneic hematopoietic stem cell transplantation (HSCT), T-cell-mediated graft-versus-host disease (GvHD) (Weiden, P L et al., *N. Engl. J. Med.* 1979; 300(19):1068-1073; Appelbaum, FR *Nature*, 2001; 411(6835):385-389; Porter, D L et al., *N. Engl. J. Med.* 1994; 330(2):100-106; Kolb, H J et al. *Blood*. 1995; 86(5):2041-2050; Slavin, S. et al., *Blood*. 1996; 87(6):2195-2204), and donor natural killer (NK) cell alloreactivity (Ruggeri L, et al. *Science*. 2002; 295(5562):2097-2100; Giebel S, et al. *Blood*. 2003; 102(3):814-819; Cooley S, et al. *Blood*. 2010; 116(14):2411-2419) are inversely related to leukemia recurrence. Besides the HSCT context, administration of antibodies that release T cells from inhibitory signals (Sharma P, et al., *Nat Rev Cancer*. 2011; 11(11):805-812; Pardoll D M., *Nat Rev Cancer*. 2012; 12(4):252-264), or bridge them to tumor cells (Topp M S, et al. *J. Clin. Oncol.* 2011; 29(18):2493-2498) produced major clinical responses in patients with either solid tumors or leukemia. Finally, infusion of genetically-modified autologous T lymphocytes induced complete and durable remission in patients with refractory leukemia and lymphoma (Maude S L, et al. *N Engl J Med*. 2014; 371(16):1507-1517).

(3) Nevertheless, there is a significant need for improving immune cell therapy by broadening its applicability and enhancing its efficacy.

SUMMARY OF THE INVENTION

(4) The present invention relates to engineered immune cells having enhanced therapeutic efficacy for, e.g., cancer therapy. In certain embodiments, the present invention provides an engineered immune cell that comprises a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule linked to a localizing domain.

(5) In other embodiments, the present invention provides the use of an engineered immune cell that comprises a gene encoding an immune activating receptor, and a gene encoding a target-binding molecule linked to a localizing domain for treating cancer, comprising administering a therapeutic amount of the engineered immune cell to a subject in need thereof.

(6) In various embodiments, the present invention also provides a method for producing an engineered immune cell, the method comprising introducing into an immune cell a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule linked to a localizing domain, thereby producing an engineered immune cell.

(7) In some embodiments, the engineered immune cells possess enhanced therapeutic efficacy as a result of one or more of reduced graft-versus-host disease (GvHD) in a host, reduced or elimination of rejection by a host, extended survival in a host, reduced inhibition by the tumor in a host,

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing will be apparent from the following more particular description of example embodiments of the invention, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments of the present invention.

(2) FIGS. 1A-1B is a schematic representation of a strategy employed in the present invention. FIG. 1A is an overall mechanism of CAR mediated killing of cancer cells. FIG. 1B shows the combined expression of CAR with different formats of compartment-directed scFv (an example of a target-binding molecule linked to a localizing domain) and examples of possible targets. The CAR can be replaced by other receptors that can enhance immune cell capacity.

(3) FIG. 2 is a schematic diagram of constructs containing scFv together with domains that localize them to specific cellular compartments. Abbreviations: β 2M, β -2 microglobulin; SP, signal peptide; VL, variable light chain; VH, variable heavy chain; TM, transmembrane domain; HA, human influenza hemagglutinin. Additional constructs not listed in the figure include membrane-bound (mb) myc EEKMP, mb myc KKTN, mb myc YQRL, mb TGN38 cytoplasmic domain, mb myc RNIKCD, linker(20-amino acid) mb EEKMP, as well as variants of constructs without signaling peptide and with a varying number of amino acids in the CD8 transmembrane domain. The nucleotide sequence of the 10-amino acid linker is GGTGGTGGCGGCAGTGGTGGCGGTGGCTCA (SEQ ID NO: 61); the amino acid sequence is GGGSGGGGS (SEQ ID NO: 62). The nucleotide sequence of the 20-amino acid linker is GGTGGTGGCGGCAGTGGTGGCGGTGGCTCAGGCGGTGGTGGCTCCGGTGGCGGTGGCTCT (SEQ ID NO: 63); the amino acid sequence is GGGSGGGSGGGSGGGGS (SEQ ID NO: 41). Various localization domains are indicated under the heading "Localization domains," and depicts linkers in some examples, as indicated. The constructs "myc KDEL" and "PEST KDEL" show the use of more than one localization domains in a single construct.

(4) FIGS. 3A-3C show downregulation of CD3/TCR in T cells by scFv targeting of CD3 ϵ . FIG. 3A shows expression of surface CD3 ϵ in Jurkat cells, transduced with either a retroviral vector containing green fluorescent protein (GFP) only ("mock") or a vector containing GFP plus different constructs as indicated. Expression of CD3 ϵ on the cell membrane was compared to that of mock-transduced cells 1 week after transduction using an anti-CD3 antibody conjugated to allophycocyanin (BD Biosciences). All comparisons were performed after gating on GFP-positive cells. FIG. 3B depicts similar experiments performed with peripheral blood T lymphocytes expanded with anti-CD3/CD28 beads (Lifesciences). Staining was performed 1 week after transduction. FIG. 3C shows flow cytometry plots illustrating downregulation of membrane CD3 ϵ in Jurkat cells after transduction with the constructs indicated. Dashed rectangles on the upper right quadrant of each plot enclose GFP+CD3+ cells.

(5) FIG. 4 shows downregulation of CD3 ϵ and TCR $\alpha\beta$ on the cell membrane in Jurkat T cells upon transduction with anti-CD3 ϵ scFv-KDEL or -PEST, or -mb EEKMP. Membrane marker expression was measured 1 week after transduction using an anti-CD3 antibody conjugated to allophycocyanin (BD Biosciences) or an anti-TCR $\alpha\beta$ conjugated to phycoerythrin (Biolegend). Lines labeled "Control" represent labelling of mock-transduced cells. Dashed vertical line represents the upper limit of staining obtained with an isotype-matched non-reactive antibody.

(6) FIG. 5 shows that anti-scFv and CAR can be expressed simultaneously. Flow cytometric dot plots represent staining of Jurkat cells (top row) or peripheral blood lymphocytes (bottom row) with anti-CD3 allophycocyanin antibody and goat-anti-mouse Fab2 biotin plus streptavidin conjugated to phycoerythrin (to detect the CAR). Cells were transduced with the anti-CD3 scFv-myc KDEL construct, the anti-CD19-4-1BB-CD3 ζ construct, or both. After gating on GFP-positive cells, those transduced with anti-CD3 scFv-myc KDEL downregulated CD3 (left column, bottom left quadrants) and those transduced with the anti-CD19-4-1BB-CD3 ζ construct expressed the CAR (middle column, top right quadrants). A substantial proportion of cells transduced with both constructs were CD3-negative and CAR-positive (right column, top left quadrants).

(7) FIG. 6 illustrates that anti-CD19 CAR triggers T cell activation and degranulation regardless of CD3/TCR downregulation. Jurkat cells were transduced with the anti-CD3 scFv-myc KDEL construct, the anti-CD19-4-1BB-CD3 ζ construct, or both. T cell activation and degranulation was compared to that of mock-transduced cells. Cells were co-cultured alone or with the CD19+ leukemia cell line OP-1 at a 1:1 ratio. After 18 hours, expression of CD69 and CD25 was tested by flow cytometry using specific antibodies (from BD Biosciences); expression of CD107a was tested after 6 hours (antibody from BD Biosciences). In the presence of OP-1 cells, CD69 and CD25 expression in CAR-expressing cells occurred regardless of whether cells were also transduced with anti-CD3 scFv-KDEL; no activation occurred in mock- or anti-CD3 scFv-myc KDEL transduced cells, or in the absence of OP-1 cells. CAR stimulation enhanced CD107 expression which was not affected by CD3/TCR downregulation.

(8) FIG. 7 shows that anti-CD19 CAR expressed in T cells causes T cell proliferation regardless of CD3/TCR downregulation. Peripheral blood T lymphocytes were transduced with both the anti-CD3 scFv-myc KDEL construct and the anti-CD19-4-1BB-CD3 ζ construct. Transduced T lymphocytes were co-cultured with OP-1 cells treated with Streck (Omaha, NE) to inhibit their proliferation for the time indicated. Expansion of CD3-positive and CD3-negative T lymphocytes expressing the anti-CD19 CAR was compared to that of mock-transduced T cells. Each symbol shows the average cell count of two parallel cultures. CAR T cell expanded equally well regardless of CD3/TCR expression.

(9) FIG. 8 shows expression of CD7 on the membrane of peripheral blood T lymphocytes transduced with either a retroviral vector containing GFP only ("mock") or a vector containing GFP plus anti-CD7 scFv-myc KDEL construct. Expression of CD7 on the cell membrane was compared to that of mock-transduced cells 1 week after transduction using an anti-CD7 antibody conjugated to phycoerythrin (BD Biosciences). Dashed rectangles on the upper right quadrant of each plot enclose GFP+CD7+ cells.

(10) FIG. 9 depicts the downregulation of HLA Class I in T cells by scFv targeting of β 2-microglobulin. Jurkat T cells were transduced with anti- β 2M scFv-myc KDEL. Expression of HLA-ABC on the cell membrane was compared to that of mock-transduced cells 1 week after transduction using an anti-HLA-ABC antibody conjugated to phycoerythrin (BD Biosciences). Staining with an isotype-matched control antibody is also shown. Analysis was performed after gating on GFP-positive cells.

(11) FIG. 10 depicts the downregulation of Killer Immunoglobulin-like Receptor (KIR) 2DL1 and KIR2DL2/DL3 in human NK cells by scFv targeting of KIR2DL1 and KIR2DL2/DL3. NK cells, expanded ex vivo and selected for KIR2DL1 expression, were transduced with anti-KIR2DL1-KIR2DL2/DL3 scFv-linker (20) AEKDEL or -EEKMP. Expression of the corresponding KIR on the cell membrane was compared to that of mock-transduced cells 8 days after transduction using an anti-KIR2DL1 antibody conjugated to allophycocyanin (R&D Systems) or an anti-KIR2DL2/DL3 antibody conjugated to phycoerythrin (BD Biosciences). Staining with an isotype-matched control antibody is also shown. Analysis was performed after gating on GFP-positive cells.

(12) FIG. 11 depicts the downregulation of NKG2A in human NK cells by scFv targeting. NK cells, expanded ex vivo, were transduced with anti-NKG2A scFv-EEKMP. Expression of NKG2A on the cell membrane was compared to that of mock-transduced cells 8 days after transduction using an NKG2A antibody conjugated to phycoerythrin (Beckman Coulter). Staining with an isotype-matched control antibody is also shown. Analysis was performed after gating on GFP-positive cells.

DETAILED DESCRIPTION OF THE INVENTION

(13) A description of example embodiments of the invention follows.

(14) In recent years, gains in knowledge about the molecular pathways that regulate immune cells have been paralleled by a remarkable evolution in the capacity to manipulate them ex vivo, including their expansion and genetic engineering. It is now possible to reliably prepare highly sophisticated

clinical-grade immune cell products in a timely fashion. A prime example of how the anti-cancer activity of immune cells can be directed and magnified by ex vivo cell engineering is the development of chimeric antigen receptor (CAR) T cells (Eshhar, Z. et al., *PNAS*. 1993; 90(2):720-724).

(15) CARs are artificial multi-molecular proteins, which have been previously described (Geiger T L, et al., *J Immunol*. 1999; 162(10):5931-5939; Brentjens R J, et al., *Nat Med*. 2003; 9(3):279-286; Cooper L J, et al., *Blood*. 2003; 101(4):1637-1644). CARs comprise an extracellular domain that binds to a specific target, a transmembrane domain, and a cytoplasmic domain. The extracellular domain and transmembrane domain can be derived from any desired source for such domains, as described in, e.g., U.S. Pat. No. 8,399,645, incorporated by reference herein in its entirety. Briefly, a CAR may be designed to contain a single-chain variable region (scFv) of an antibody that binds specifically to a target. The scFv may be linked to a T-cell receptor (TCR)-associated signaling molecule, such as CD3 ζ , via transmembrane and hinge domains. Ligation of scFv to the cognate antigen triggers signal transduction. Thus, CARs can instantaneously redirect cytotoxic T lymphocytes towards cancer cells and provoke tumor cell lysis (Eshhar, Z. et al., *PNAS*. 1993; 90(2):720-724; Geiger T L, et al., *J Immunol*. 1999; 162(10):5931-5939; Brentjens R J, et al., *Nat Med*. 2003; 9(3):279-286; Cooper L J, et al., *Blood*. 2003; 101(4):1637-1644; Imai C, et al., *Leukemia*. 2004; 18:676-684). Because CD3 ζ signaling alone is not sufficient to durably activate T cells (Schwartz R H. *Annu Rev Immunol*. 2003; 21:305-334; Zang X and Allison J P. *Clin Cancer Res*. 2007; 13(18 Pt 1):5271-5279), co-stimulatory molecules such as CD28 and 4-1BB (or CD137) have been incorporated into CAR constructs to boost signal transduction. This dual signaling design ("second generation CAR") is useful to elicit effective anti-tumor activity from T cells (Imai C, et al., *Leukemia*. 2004; 18:676-684; Campana D, et al., *Cancer J*. 2014; 20(2):134-140).

(16) A specific CAR, anti-CD19 CAR, containing both 4-1BB and CD3 ζ has been described in U.S. Pat. No. 8,399,645. Infusion of autologous T cells expressing an anti-CD19-4-1BB-CD3 ζ CAR resulted in dramatic clinical responses in patients with chronic lymphocytic leukemia (CLL) (Porter D L, et al., Chimeric antigen receptor-modified T cells in chronic lymphoid leukemia; 2011: *N Engl J Med*. 2011; 365(8):725-733; Kalos M, et al., *Sci Transl Med*. 2011; 3(95):95ra73), and acute lymphoblastic leukemia (ALL) (Grupp S A, et al., *N Engl J Med*. 2013; 368(16):1509-1518; Maude S L, et al., *N Engl J Med*. 2014; 371(16):1507-1517). These studies, and studies with CARs bearing different signaling modules (Till B G, et al., *Blood*. 2012; 119(17):3940-3950; Kochenderfer J N, et al., *Blood*. 2012; 119(12):2709-2720; Brentjens R J, et al., *Blood*. 2011; 118(18):4817-4828; Brentjens R J, et al., *Sci Transl Med*. 2013; 5(177):177ra138), provide a convincing demonstration of the clinical potential of this technology, and of immunotherapy in general.

(17) The methods described herein enable rapid removal or inactivation of specific proteins in immune cells redirected by a natural or artificial receptor, e.g., CARs, thus broadening the application potential and significantly improving the function of the engineered cells. The method relies, in part, on a single construct or multiple constructs containing an immune activating receptor, e.g., a CAR (which comprises an extracellular domain (e.g., an scFv) that binds to a specific target, a transmembrane domain, and a cytoplasmic domain) together with a target-binding molecule that binds a target (e.g., protein) to be removed or neutralized; the target-binding molecule is linked to a domain (i.e., localizing domain) that directs it to specific cellular compartments, such as the Golgi or endoplasmic reticulum, the proteasome, or the cell membrane, depending on the application. For simplicity, a target-binding molecule linked to a localizing domain (LD) is sometimes referred to herein as "LD-linked target-binding molecule."

(18) As will be apparent from the teachings herein, a variety of immune activating receptors may be suitable for the methods of the present invention. That is, any receptor that comprises a molecule that, upon binding (ligation) to a ligand (e.g., peptide or antigen) expressed on a cancer cell, is capable of activating an immune response may be used according to the present methods. For example, as described above, the immune activating receptor can be a chimeric antigen receptor (CAR); methods for designing and manipulating a CAR is known in the art (see, Geiger T L, et al., *J Immunol*. 1999; 162(10):5931-5939; Brentjens R J, et al., *Nat Med*. 2003; 9(3):279-286; Cooper L J, et al., *Blood*. 2003; 101(4):1637-1644). Additionally, receptors with antibody-binding capacity can be used (e.g., CD16-4-1BB-CD3zeta receptor—Kudo K, et al. *Cancer Res*. 2014; 74(1):93-103), which are similar to CARs, but with the scFv replaced with an antibody-binding molecule (e.g., CD16, CD64, CD32). Further, T-cell receptors comprising T-cell receptor alpha and beta chains that bind to a peptide expressed on a tumor cell in the context of the tumor cell HLA can also be used according to the present methods. In addition, other receptors bearing molecules that activate an immune response by binding a ligand expressed on a cancer cell can also be used—e.g. NKG2D-DAP10-CD3zeta receptor, which binds to NKG2D ligand expressed on tumor cells (see, e.g., Chang Y H, et al., *Cancer Res*. 2013; 73(6):1777-1786). All such suitable receptors collectively, as used herein, are referred to as an "immune activating receptor" or a "receptor that activates an immune response upon binding a cancer cell ligand." Therefore, an immune activating receptor having a molecule activated by a cancer cell ligand can be expressed together with a LD-linked target-binding molecule according to the present methods.

(19) The present methods significantly expand the potential applications of immunotherapies based on the infusion of immune cells redirected by artificial receptors. The method described is practical and can be easily incorporated in a clinical-grade cell processing. For example, a single bicistronic construct containing, e.g., a CAR and a LD-linked target-binding molecule, e.g., scFv-myc KDEL (or PEST or transmembrane) can be prepared by inserting an internal ribosomal entry site (IRES) or a 2A peptide-coding region site between the 2 cDNAs encoding the CAR and the LD-linked target-binding molecule. The design of tricistronic delivery systems to delete more than one target should also be feasible. Alternatively, separate transductions of the 2 genes (simultaneously or sequentially) could be performed. In the context of cancer cell therapy, the CAR could be replaced by an antibody-binding signaling receptor (Kudo K, et al., *Cancer Res*. 2014; 74(1):93-103), a T-cell receptor directed against a specific HLA-peptide combination, or any receptor activated by contact with cancer cells (Chang Y H, et al., *Cancer Res*. 2013; 73(6):1777-1786). The results of the studies described herein with simultaneous anti-CD19-4-1BB-CD3 ζ CAR and anti-CD3 ϵ scFv-KDEL demonstrate that the signaling capacity of the CAR was not impaired.

(20) Both the anti-CD3 ϵ scFv-KDEL (and -PEST) tested herein stably downregulate CD3 as well as TCR expression. Residual CD3 $^{+}$ T cells could be removed using CD3 beads, an approach that is also available in a clinical-grade format. The capacity to generate CD3/TCR-negative cells that respond to CAR signaling represents an important advance. Clinical studies with CAR T cells have generally been performed using autologous T cells. Thus, the quality of the cell product varies from patient to patient and responses are heterogeneous. Infusion of allogeneic T cells is currently impossible as it has an unacceptably high risk of potentially fatal GvHD, due to the stimulation of the endogenous TCR by the recipient's tissue antigens. Downregulation of CD3/TCR opens the possibility of infusing allogeneic T cells because lack of endogenous TCR eliminates GvHD capacity. Allogeneic products could be prepared with the optimal cellular composition (e.g., enriched in highly cytotoxic T cells, depleted of regulatory T cells, etc.) and selected so that the cells infused have high CAR expression and functional potency. Moreover, fully standardized products could be cryopreserved and be available for use regardless of the patient immune cell status and his/her fitness to undergo apheresis or extensive blood draws. Removal of TCR expression has been addressed using gene editing tools, such as nucleases (Torikai H, et al. *Blood*, 2012; 119(24):5697-5705). Although this is an effective approach, it is difficult to implement in a clinical setting as it requires several rounds of cell selection and expansion, with prolonged culture. The methods described herein have considerable practical advantages.

(21) Additionally, a LD-linked target-binding molecule (e.g., scFv-myc KDEL, scFv-EEKKMP or scFv-PEST, wherein scFv targets a specific protein/molecule) can be used according to the present invention to delete HLA Class I molecules, reducing the possibility of rejection of allogeneic cells. While infusion of allogeneic T cells is a future goal of CAR T cell therapy, infusion of allogeneic natural killer (NK) cells is already in use to treat patients with cancer. A key factor that determines the success of NK cell-based therapy is that NK cells must persist in sufficient numbers to achieve an effector:target ratio likely to produce tumor cytoreduction (Miller J S. *Hematology Am Soc Hematol Educ Program*. 2013; 2013:247-253). However, when allogeneic cells are infused, their persistence is limited. Immunosuppressive chemotherapy given to the patient allows transient engraftment of the infused NK cells but these are rejected within 2-4 weeks of infusion (Miller J S, et al. *Blood*. 2005; 105:3051-3057; Rubnitz J E, et al., *J Clin Oncol*. 2010; 28(6):955-959). Contrary to organ transplantation, continuing immunosuppression is not an option because immunosuppressive drugs also suppress NK cell function. Because rejection is primarily mediated by recognition of HLA Class I molecules by the

recipient CD8⁺T lymphocytes, removing HLA Class I molecules from the infused NK cells (or T cells) will diminish or abrogate the rejection rate, extend the survival of allogeneic cells, and hence their anti-tumor capacity.

(22) Furthermore, a LD-linked target-binding molecule can be used according to the present invention to target inhibitory receptors. Specifically, administration of antibodies that release T cells from inhibitory signals such as anti-PD1 or anti-CTLA-4 have produced dramatic clinical responses (Sharma P, et al., *Nat Rev Cancer*. 2011; 11(11):805-812; Pardoll D M. *Nat Rev Cancer*. 2012; 12(4):252-264). CAR-T cells, particularly those directed against solid tumors, might be inhibited by similar mechanisms. Thus, expression of a target-binding molecule (e.g., scFv or ligands) against PD1, CTLA-4, Tim3 or other inhibitory receptors would prevent the expression of these molecules (if linked to, e.g., KDEL (SEQ ID NO: 4), EEKKMP (SEQ ID NO: 64) or PEST motif SHGFPPEVEEQDDGTLPMSCAQESGMDRHPAACASARINV (SEQ ID NO: 7)) or prevent binding of the receptors to their ligands (if linked to a transmembrane domain) and sustain CAR-mediated signal transduction. In NK cells, examples of inhibitory receptors include killer immunoglobulin-like receptors (KIRs) and NKG2A (Vivier E, et al., *Science*, 2011; 331(6013):44-49).

(23) The methods of the present invention also enable targeting of a greater number of targets amenable for CAR-directed T cell therapy. One of the main limitations of CAR-directed therapy is the paucity of specific antigens expressed by tumor cells. In the case of hematologic malignancies, such as leukemias and lymphomas, molecules which are not expressed in non-hematopoietic cells could be potential targets but cannot be used as CAR targets because they are also expressed on T cells and/or NK cells. Expressing such CARs on immune cells would likely lead to the demise of the immune cells themselves by a “fratricidal” mechanism, nullifying their anti-cancer capacity. If the target molecule can be removed from immune cells without adverse functional effects, then the CAR with the corresponding specificity can be expressed. This opens many new opportunities to target hematologic malignancies. Examples of the possible targets include CD38 expressed in multiple myeloma, CD7 expressed in T cell leukemia and lymphoma, Tim-3 expressed in acute leukemia, CD30 expressed in Hodgkin disease, CD45 and CD52 expressed in all hematologic malignancies. These molecules are also expressed in a substantial proportion of T cells and NK cells.

(24) Moreover, it has been shown that secretion of cytokines by activated immune cells triggers cytokine release syndrome and macrophage activation syndrome, presenting serious adverse effects of immune cell therapy (Lee D W, et al., *Blood*. 2014; 124(2):188-195). Thus, the LD-linked target-binding molecule can be used according to the present invention to block cytokines such as IL-6, IL-2, IL-4, IL-7, IL-10, IL-12, IL-15, IL-18, IL-21, IL-27, IL-35, interferon (IFN)- γ , IFN- β , IFN- α , tumor necrosis factor (TNF)- α , and transforming growth factor (TGF)- β , which may contribute to such inflammatory cascade.

(25) Accordingly, in one embodiment, the present invention relates to an engineered immune cell that comprises a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule linked to a localizing domain.

(26) As used herein, an “engineered” immune cell includes an immune cell that has been genetically modified as compared to a naturally-occurring immune cell. For example, an engineered T cell produced according to the present methods carries a nucleic acid comprising a nucleotide sequence that does not naturally occur in a T cell from which it was derived. In some embodiments, the engineered immune cell of the present invention includes a chimeric antigen receptor (CAR) and a target-binding molecule linked to a localizing domain (LD-linked target-binding molecule). In a particular embodiment, the engineered immune cell of the present invention includes an anti-CD19-4-1BB-CD3 ζ CAR and an anti-CD3 scFv linked to a localizing domain.

(27) In certain embodiments, the engineered immune cell is an engineered T cell, an engineered natural killer (NK) cell, an engineered NK/T cell, an engineered monocyte, an engineered macrophage, or an engineered dendritic cell.

(28) In certain embodiments, an “immune activating receptor” as used herein refers to a receptor that activates an immune response upon binding a cancer cell ligand. In some embodiments, the immune activating receptor comprises a molecule that, upon binding (ligation) to a ligand (e.g., peptide or antigen) expressed on a cancer cell, is capable of activating an immune response. In one embodiment, the immune activating receptor is a chimeric antigen receptor (CAR); methods for designing and manipulating a CAR are known in the art. In other embodiments, the immune activating receptor is an antibody-binding receptor, which is similar to a CAR, but with the scFv replaced with an antibody-binding molecule (e.g., CD16, CD64, CD32) (see e.g., CD16-4-1BB-CD3zeta receptor—Kudo K, et al. *Cancer Res*. 2014; 74(1):93-103). In various embodiments, T-cell receptors comprising T-cell receptor alpha and beta chains that bind to a peptide expressed on a tumor cell in the context of the tumor cell HLA can also be used according to the present methods. In certain embodiments, other receptors bearing molecules that activate an immune response by binding a ligand expressed on a cancer cell can also be used—e.g., NKG2D-DAP10-CD3zeta receptor, which binds to NKG2D ligand expressed on tumor cells (see, e.g., Chang Y H, et al., *Cancer Res*. 2013; 73(6):1777-1786). All such suitable receptors capable of activating an immune response upon binding (ligation) to a ligand (e.g., peptide or antigen) expressed on a cancer cell are collectively referred to as an “immune activating receptor.” As would be appreciated by those of skill in the art, an immune activating receptor need not contain an antibody or antigen-binding fragment (e.g., scFv); rather the portion of the immune activating receptor that binds to a target molecule can be derived from, e.g., a receptor in a receptor-ligand pair, or a ligand in a receptor-ligand pair.

(29) In certain aspects, the immune activating receptor binds to molecules expressed on the surface of tumor cells, including but not limited to, CD20, CD22, CD33, CD2, CD3, CD4, CD5, CD7, CD8, CD45, CD52, CD38, CS-1, TIM3, CD123, mesothelin, folate receptor, HER2-neu, epidermal-growth factor receptor, and epidermal growth factor receptor. In some embodiments, the immune activating receptor is a CAR (e.g., anti-CD19-4-1BB-CD3 ζ CAR). In certain embodiments, the immune activating receptor comprises an antibody or antigen-binding fragment thereof (e.g., scFv) that binds to molecules expressed on the surface of tumor cells, including but not limited to, CD20, CD22, CD33, CD2, CD3, CD4, CD5, CD7, CD8, CD45, CD52, CD38, CS-1, TIM3, CD123, mesothelin, folate receptor, HER2-neu, epidermal-growth factor receptor, and epidermal growth factor receptor. Antibodies to such molecules expressed on the surface of tumor cells are known and available in the art. By way of example, antibodies to CD3 and CD7 are commercially available and known in the art. Such antibodies, as well as fragments of antibodies (e.g., scFv) derived therefrom, can be used in the present invention, as exemplified herein. Further, methods of producing antibodies and antibody fragments against a target protein are well-known and routine in the art.

(30) The transmembrane domain of an immune activating receptor according to the present invention (e.g., CAR) can be derived from a single-pass membrane protein, including, but not limited to, CD8 α , CD8 β , 4-1BB, CD28, CD34, CD4, Fc ϵ R1 γ , CD16 (e.g., CD16A or CD16B), OX40, CD3 ζ , CD38, CD3 γ , CD3 δ , TCR α , CD32 (e.g., CD32A or CD32B), CD64 (e.g., CD64A, CD64B, or CD64C), VEGFR2, FAS, and FGFR2B. In some examples, the membrane protein is not CD8 α . The transmembrane domain may also be a non-naturally occurring hydrophobic protein segment.

(31) The hinge domain of the immune activating receptor (e.g., CAR) can be derived from a protein such as CD8 α , or IgG. The hinge domain can be a fragment of the transmembrane or hinge domain of CD8 α , or a non-naturally occurring peptide, such as a polypeptide consisting of hydrophilic residues of varying length, or a (GGGG).sub.n (SEQ ID NO: 8) polypeptide, in which n is an integer of, e.g., 3-12, inclusive.

(32) The signaling domain of the immune activating receptor (e.g., CAR) can be derived from CD3 ζ , Fc ϵ R1 γ , DAP10, DAP12 or other molecules known to deliver activating signals in immune cells. At least one co-stimulatory signaling domain of the receptor can be a co-stimulatory molecule such as 4-1BB (also known as CD137), CD28, CD28.sub.LL.fwdarw.GG variant, OX40, ICOS, CD27, GITR, HVEM, TIM1, LFA1, or CD2. Such molecules are readily available and known in the art.

(33) As would be appreciated by those of skill in the art, the components of an immune activating receptor can be engineered to comprise a number of functional combinations, as described herein, to produce a desired result. Using the particular CAR anti-CD19-4-1BB-CD3 ζ as an example, the antibody (e.g., or antigen-binding fragment thereof such as an scFv) that binds a molecule can be substituted for an antibody that binds different molecule, as described herein (e.g., anti-CD20, anti-CD33, anti-CD123, etc., instead of anti-CD19). In other embodiments, the co-stimulatory molecule (4-1BB in this specific example) can also be varied with a different co-stimulatory molecule, e.g., CD28. In some embodiments, the

stimulatory molecule (CD3 ζ in this specific example), can be substituted with another known stimulatory molecule. In various embodiments, the transmembrane domain of the receptor can also be varied as desired. The design, production, and testing for functionality of such immune activating receptors can be readily determined by those of skill in the art. Similarly, the design, delivery into cells and expression of nucleic acids encoding such immune activating receptors are readily known and available in the art.

(34) As used herein, the term “nucleic acid” refers to a polymer comprising multiple nucleotide monomers (e.g., ribonucleotide monomers or deoxyribonucleotide monomers). “Nucleic acid” includes, for example, genomic DNA, cDNA, RNA, and DNA-RNA hybrid molecules. Nucleic acid molecules can be naturally occurring, recombinant, or synthetic. In addition, nucleic acid molecules can be single-stranded, double-stranded or triple-stranded. In some embodiments, nucleic acid molecules can be modified. In the case of a double-stranded polymer, “nucleic acid” can refer to either or both strands of the molecule.

(35) The term “nucleotide sequence,” in reference to a nucleic acid, refers to a contiguous series of nucleotides that are joined by covalent linkages, such as phosphorus linkages (e.g., phosphodiester, alkyl and aryl-phosphonate, phosphorothioate, phosphotriester bonds), and/or non-phosphorus linkages (e.g., peptide and/or sulfamate bonds). In certain embodiments, the nucleotide sequence encoding, e.g., a target-binding molecule linked to a localizing domain is a heterologous sequence (e.g., a gene that is of a different species or cell type origin).

(36) The terms “nucleotide” and “nucleotide monomer” refer to naturally occurring ribonucleotide or deoxyribonucleotide monomers, as well as non-naturally occurring derivatives and analogs thereof. Accordingly, nucleotides can include, for example, nucleotides comprising naturally occurring bases (e.g., adenosine, thymidine, guanosine, cytidine, uridine, inosine, deoxyadenosine, deoxythymidine, deoxyguanosine, or deoxycytidine) and nucleotides comprising modified bases known in the art.

(37) As will be appreciated by those of skill in the art, in some aspects, the nucleic acid further comprises a plasmid sequence. The plasmid sequence can include, for example, one or more sequences selected from the group consisting of a promoter sequence, a selection marker sequence, and a locus-targeting sequence.

(38) As used herein, the gene encoding a target-binding molecule linked to a localizing domain is sometimes referred to as “LD-linked target-binding molecule.”

(39) In certain embodiments, the target-binding molecule is an antibody or antigen-binding fragment thereof. As used herein, “antibody” means an intact antibody or antigen-binding fragment of an antibody, including an intact antibody or antigen-binding fragment that has been modified or engineered, or that is a human antibody. Examples of antibodies that have been modified or engineered are chimeric antibodies, humanized antibodies, multiparatopic antibodies (e.g., biparatopic antibodies), and multispecific antibodies (e.g., bispecific antibodies). Examples of antigen-binding fragments include Fab, Fab', F(ab').sub.2, Fv, single chain antibodies (e.g., scFv), minibodies and diabodies.

(40) A “Fab fragment” comprises one light chain and the C.sub.H1 and variable regions of one heavy chain. The heavy chain of a Fab molecule cannot form a disulfide bond with another heavy chain molecule.

(41) An “Fc” region contains two heavy chain fragments comprising the CH2 and CH3 domains of an antibody. The two heavy chain fragments are held together by two or more disulfide bonds and by hydrophobic interactions of the CH3 domains.

(42) A “Fab' fragment” contains one light chain and a portion of one heavy chain that contains the VH domain and the CH1 domain and also the region between the CH1 and CH2 domains, such that an interchain disulfide bond can be formed between the two heavy chains of two Fab' fragments to form a F(ab').sub.2 molecule.

(43) A “F(ab').sub.2 fragment” contains two light chains and two heavy chains containing a portion of the constant region between the C.sub.H1 and C.sub.H.sub.2 domains, such that an interchain disulfide bond is formed between the two heavy chains. A F(ab').sub.2 fragment thus is composed of two Fab' fragments that are held together by a disulfide bond between the two heavy chains.

(44) The “Fv region” comprises the variable regions from both the heavy and light chains, but lacks the constant regions.

(45) In a particular embodiment, the target-binding molecule is single-chain Fv antibody (“scFv antibody”). scFv refers to antibody fragments comprising the VH and VL domains of an antibody, wherein these domains are present in a single polypeptide chain. Generally, the Fv polypeptide further comprises a polypeptide linker between the VH and VL domains which enables the scFv to form the desired structure for antigen binding. For a review of scFv, see Pluckthun (1994) *The Pharmacology Of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds. Springer-Verlag, New York, pp. 269-315. See also, PCT Publication No. WO 88/01649 and U.S. Pat. Nos. 4,946,778 and 5,260,203. By way of example, the linker between the VH and VL domains of the scFvs disclosed herein comprise, e.g., GGGSGGGSGGGSGGGSGGGGS (SEQ ID NO: 41) or GGGSGGGSGGGSGGGGS (SEQ ID NO: 43). As would be appreciated by those of skill in the art, various suitable linkers can be designed and tested for optimal function, as provided in the art, and as disclosed herein.

(46) The scFv that is part of the LD-linked target-binding molecule is not necessarily the same as the scFv that occurs in the context of, e.g., a chimeric antigen receptor (CAR) or a similar antibody-binding signaling receptor. In some embodiments, the scFv that is part of the LD-linked target-binding molecule is the same as the scFv that occurs in the context of, e.g., a chimeric antigen receptor (CAR) or a similar antibody-binding signaling receptor.

(47) In some embodiments, the nucleic acid comprising a nucleotide sequence encoding a target-binding molecule (e.g., an scFv in the context of a LD-linked target-binding molecule) comprises one or more sequences that have at least 80%, at least 85%, at least 88%, at least 90%, at least 92%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any one or more of SEQ ID NOs: 14, 15, 18, 19, 22, 23, 26, 27, 30, 31, 34, 35, 38, or 39.

(48) The term “sequence identity” means that two nucleotide or amino acid sequences, when optimally aligned, such as by the programs GAP or BESTFIT using default gap weights, share at least, e.g., 70% sequence identity, or at least 80% sequence identity, or at least 85% sequence identity, or at least 90% sequence identity, or at least 95% sequence identity or more. For sequence comparison, typically one sequence acts as a reference sequence (e.g., parent sequence), to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

(49) Optimal alignment of sequences for comparison can be conducted, e.g., by the local homology algorithm of Smith & Waterman, *Adv. Appl. Math.* 2:482 (1981), by the homology alignment algorithm of Needleman & Wunsch, *J Mol. Biol.* 48:443 (1970), by the search for similarity method of Pearson & Lipman, *Proc. Nat'l. Acad. Sci. USA* 85:2444 (1988), by computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, Wis.), or by visual inspection (see generally Ausubel et al., *Current Protocols in Molecular Biology*). One example of algorithm that is suitable for determining percent sequence identity and sequence similarity is the BLAST algorithm, which is described in Altschul et al., *J. Mol. Biol.* 215:403 (1990). Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (publicly accessible through the National Institutes of Health NCBI internet server). Typically, default program parameters can be used to perform the sequence comparison, although customized parameters can also be used. For amino acid sequences, the BLASTP program uses as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff & Henikoff, *Proc. Natl. Acad. Sci. USA* 89:10915 (1989)).

(50) In certain embodiments, the antibody (e.g., scFv) comprises VH and VL having amino acid sequences set forth in SEQ ID NO: 12 and 13, respectively; SEQ ID NO: 16 and 17, respectively; SEQ ID NO: 20 and 21, respectively; SEQ ID NO: 24 and 25, respectively; SEQ ID NO: 28 and 29, respectively; SEQ ID NO: 32 and 33, respectively; or SEQ ID NO: 36 and 37, respectively. In some embodiments, the antibody (e.g., scFv) comprises VH and VL having sequence that each have at least 90% sequence identity, at least 91% sequence identity, at least 92% sequence identity,

at least 93% sequence identity, at least 94% sequence identity, at least 95% sequence identity, at least 96% sequence identity, at least 97% sequence identity, at least 98% sequence identity, at least 99% sequence identity, or 100% sequence identity to the VH and VL sequences set forth in SEQ ID NO: 12 and 13, respectively; SEQ ID NO: 16 and 17, respectively; SEQ ID NO: 20 and 21, respectively; SEQ ID NO: 24 and 25, respectively; SEQ ID NO: 28 and 29, respectively; SEQ ID NO: 32 and 33, respectively; or SEQ ID NO: 36 and 37, respectively.

(51) A “diabody” is a small antibody fragment with two antigen-binding sites. The fragments comprise a heavy chain variable region (VH) connected to a light chain variable region (VL) in the same polypeptide chain (VH-VL or VL-VH). By using a linker that is too short to allow pairing between the two domains on the same chain, the domains are forced to pair with the complementary domains of another chain and create two antigen-binding sites. Diabodies are described in, e.g., patent documents EP 404,097; WO 93/11161; and Holliger et al., (1993) *Proc. Nat. Acad. Sci. USA* 90: 6444-6448.

(52) In certain embodiments, the antibody is a triabody or a tetrabody. Methods of designing and producing triabodies and tetrabodies are known in the art. See, e.g. Todorovska et al., *J. Immunol. Methods* 248(1-2):47-66, 2001.

(53) A “domain antibody fragment” is an immunologically functional immunoglobulin fragment containing only the variable region of a heavy chain or the variable region of a light chain. In some instances, two or more VH regions are covalently joined with a peptide linker to create a bivalent domain antibody fragment. The two VH regions of a bivalent domain antibody fragment may target the same or different antigens.

(54) In some embodiments, the antibody is modified or engineered. Examples of modified or engineered antibodies include chimeric antibodies, multiparatopic antibodies (e.g., biparatopic antibodies), and multispecific antibodies (e.g., bispecific antibodies).

(55) As used herein, “multiparatopic antibody” means an antibody that comprises at least two single domain antibodies, in which at least one single domain antibody is directed against a first antigenic determinant on an antigen and at least one other single domain antibody is directed against a second antigenic determinant on the same antigen. Thus, for example, a “biparatopic” antibody comprises at least one single domain antibody directed against a first antigenic determinant on an antigen and at least one further single domain antibody directed against a second antigenic determinant on the same antigen.

(56) As used herein, “multispecific antibody” means an antibody that comprises at least two single domain antibodies, in which at least one single domain antibody is directed against a first antigen and at least one other single domain antibody is directed against a second antigen (different from the first antigen). Thus, for example, a “bispecific” antibody is one that comprises at least one single domain antibody directed against a first antigen and at least one further single domain antibody directed against a second antigen, e.g., different from the first antigen.

(57) In some embodiments, the antibodies disclosed herein are monoclonal antibodies, e.g., murine monoclonal antibodies. Methods of producing monoclonal antibodies are known in the art. See, for example, Pluckthun (1994) *The Pharmacology of Monoclonal Antibodies*, Vol. 113, Rosenberg and Moore eds. Springer-Verlag, New York, pp. 269-315.

(58) In various embodiments, the target-binding molecule in the context of a LD-linked target-binding molecule is a receptor or a ligand that binds to a target molecule. For example, that target-binding molecule can be a ligand that binds PD-1 (e.g., PD-L1 or PD-L2). Thus, as would be appreciated by those of skill in the art, the target-binding molecule can be an antibody, or a ligand/receptor that binds a target molecule.

(59) As used herein, “linked” in the context of a LD-linked target-binding molecule refers to a gene encoding a target-binding molecule directly in frame (e.g., without a linker) adjacent to one or more genes encoding one or more localizing domains. Alternatively, the gene encoding a target-binding molecule may be connected to one or more gene encoding one or more localizing domains through a linker sequence, as described herein. Various suitable linkers known in the art can be used to tether the target-binding molecule to a localizing domain. For example, non-naturally occurring peptides, such as a polypeptide consisting of hydrophilic residues of varying length, or a (GGGS)_n (SEQ ID NO: 8) polypeptide, in which n is an integer of, e.g., 3-12, inclusive, can be used according to the present invention. In particular embodiments, the linker comprises, e.g., GGGSGGGSGGS (SEQ ID NO: 62). In some embodiments, the linker comprises, e.g., GGGSGGGSGGGSGGGSGGS (SEQ ID NO: 41). In various embodiments, peptide linkers having lengths of about 5 to about 100 amino acids, inclusive, can be used in the present invention. In certain embodiments, peptide linkers having lengths of about 20 to about 40 amino acids, inclusive, can be used in the present invention. In some embodiments, peptide linkers having lengths of at least 5 amino acids, at least 10 amino acids, at least 15 amino acids, at least 20 amino acids, at least 25 amino acids, at least 30 amino acids, at least 35 amino acids, or at least 40 amino acids can be used in the present invention. As would be appreciated by those of skill in the art, such linker sequences as well as variants of such linker sequences are known in the art. Methods of designing constructs that incorporate linker sequences as well as methods of assessing functionality are readily available to those of skill in the art.

(60) In certain embodiments, the LD-linked target-binding molecule binds to a target expressed on the surface of an immune cell. In some embodiments, the LD-linked target-binding molecule inhibits the activity or function of the target molecule. By way of example, as disclosed herein, the LD-linked target-binding molecule can be designed to bind to, e.g. CD3, CD7, CD45, hB2MG, KIR2DL1, KIR2DL2/DL3, or NKG2A, thereby downregulating the cell surface expression of such molecules. Downregulation of such molecules can be achieved through, for example, localizing/targeting the molecules for degradation and/or internalization. In other embodiments, the LD-linked target-binding molecule renders the target inactive (e.g., the target can no longer interact and/or bind to its cognate ligand or receptor).

(61) In some embodiments, the engineered immune cells of the present invention have enhanced therapeutic efficacy. As used herein, “enhanced therapeutic efficacy” refers to one or more of reduced graft-versus-host disease (GvHD) in a host, reduced or elimination of rejection by a host, extended survival in a host, reduced inhibition by the tumor in a host, reduced self-killing in a host, reduced inflammatory cascade in a host, or sustained CAR-mediated signal transduction in a host.

(62) In certain embodiments of the present invention, the target-binding molecule in the context of a LD-linked target-binding molecule binds to a molecule in a CD3/T-cell receptor (TCR) complex, a cytokine, a human leukocyte antigen (HLA) Class I molecule, or a receptor that downregulates immune response.

(63) In certain embodiments, a molecule in a CD3/TCR complex can be CD3ε, TCRα, TCRβ, TCRγ, TCRδ, CD3δ, CD3γ, or CD3ζ. In a particular embodiment, the molecule is CD3ε.

(64) In another embodiment, the HLA Class I molecule is beta-2 microglobulin, α1-microglobulin, α2-microglobulin, or α3-microglobulin.

(65) In other embodiments, a receptor that downregulates immune response is selected from, e.g., PD-1, CTLA-4, Tim3, killer immunoglobulin-like receptors (KIRs)—e.g. KIR2DL1 (also known as CD158a), KIR2DL2/DL3 (also known as CD158b), CD94 or NKG2A (also known as CD159a), protein tyrosine phosphatases such as Src homology region 2 domain-containing phosphatase (SHP)-1 and SHP-2. Thus, such receptors can be targeted by moiety LD-linked target-binding molecule, as described herein.

(66) In various embodiments, examples of cytokines that can be targeted with moiety LD-linked target-binding molecule include, e.g., interleukin (IL)-6, IL-2, IL-4, IL-7, IL-10, IL-12, IL-15, IL-18, IL-21, IL-27, IL-35, interferon (IFN)-γ, IFN-β, IFN-α, tumor necrosis factor (TNF)-α, or transforming growth factor (TGF)-β.

(67) In a further aspect, the LD-linked target-binding molecule binds to a molecule selected from, e.g., CD2, CD4, CD5, CD7, CD8, CD30, CD38, CD45, CD52, or CD127.

(68) Methods of producing antibodies and antibody fragments thereof against any target protein are well-known and routine in the art. Moreover, as exemplified herein, commercially available antibodies to various targets, e.g., CD3 and CD7 can be used to generate a LD-linked target-binding molecule, as exemplified herein. Antibodies known in the art, as well as fragments of antibodies (e.g., scFv) derived therefrom, can be used in the present invention, as exemplified herein.

(69) In other aspects, the localizing domain of the LD-linked target-binding molecule comprises an endoplasmic reticulum (ER) retention sequence KDEL (SEQ ID NO: 4), or other ER or Golgi retention sequences such as KKXX (SEQ ID NO: 9), KXD/E (SEQ ID NO: 10) (where X can be any

amino acid—see Gao C, et al., *Trends in Plant Science* 19: 508-515, 2014) and YQRL (SEQ ID NO: 11) (see Zhan J, et al., *Cancer Immunol Immunother* 46:55-60, 1998); a proteosome targeting sequence that comprises, e.g., “PEST” motif—SHGFPPEVEEQDDGTLPMSCAQESGMDRHPAACASARINV (SEQ ID NO: 7); and/or a sequence that targets the target-binding molecule to the cell membrane, such as the CD8 α transmembrane domain, or the transmembrane of another single-pass membrane protein, as described herein (e.g., CD8 α , CD8 β , 4-1BB, CD28, CD34, CD4, Fc ϵ RI γ , CD16 (such as CD16A or CD16B), OX40, CD3 ζ , CD3 ϵ , CD3 γ , CD3 δ , TCR α , CD32 (such as CD32A or CD32B), CD64 (such as CD64A, CD64B, or CD64C), VEGFR2, FAS, or FGFR2B). Examples of particular localizing domains (sequences) exemplified herein are shown in FIG. 2. Various other localizing sequences are known and available in the art.

(70) As shown in FIG. 2, the LD-linked target-binding molecules of the present invention can comprise one or more localizing domains. For example, the LD-linked target-binding molecule can have at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, or at least ten localizing domains linked together. When more than one localizing domain is used in a given LD-linked target-binding molecule, each localizing domain can be linked with or without any intervening linker. By way of example, as shown in FIG. 2, localization domains CD8 TM, PEST motif, and EEKKMP can be used in a single LD-linked target-binding molecule. While this particular construct shows the localization domains without any intervening linkers, various intervening linkers can be incorporated between some or all of the localization domains. Other examples are shown in FIG. 2.

(71) As would be appreciated by those of skill in the art, the immune activating receptor and/or the LD-linked target-binding molecule can be designed to bind to the targets disclosed herein, as well as variants of the targets disclosed herein. By way of example, an immune activating receptor and/or the LD-linked target-binding molecule can be designed to bind to a molecule in a CD3/TCR complex, or a naturally-occurring variant molecule thereof. Such naturally-occurring variants can have the same function as the wild-type form of the molecule. In other embodiments, the variant can have a function that is altered relative to the wild-type form of the molecule (e.g., confers a diseased state).

(72) As would be appreciated by those of skill in the art, the various components of the LD-linked target-binding molecule constructs shown in FIG. 2 can be substituted in different combinations (e.g., to contain a different linker, different localizing sequence, different scFv, etc.), so long as the combination produces a functional LD-linked target-binding molecule. Methods of assessing functionality for a particular construct are within the ambit of those of skill in the art, as disclosed herein.

(73) In further aspects, the present invention relates to the use of an engineered immune cell that comprises a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule (e.g., scFv) linked to a localizing domain for treating cancer, comprising administering a therapeutic amount of the engineered immune cell to a subject in need thereof.

(74) In another aspect, the present invention relates to the use of an engineered immune cell that comprises a nucleic acid comprising a nucleotide sequence encoding a chimeric antigen receptor (CAR) and a nucleic acid comprising a nucleotide sequence encoding a single-chain variable fragment (scFv) linked to a localizing domain for treating cancer, comprising administering a therapeutic amount of the engineered immune cell to a subject in need thereof.

(75) In other aspects, the present invention relates to the use of an engineered immune cell that comprises a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule (e.g., scFv) linked to a localizing domain for treating an autoimmune disorder, comprising administering a therapeutic amount of the engineered immune cell to a subject in need thereof.

(76) In other aspects, the present invention also relates to the use of an engineered immune cell that comprises a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule (e.g., scFv) linked to a localizing domain for treating an infectious disease, comprising administering a therapeutic amount of the engineered immune cell to a subject in need thereof.

(77) In various embodiments, the immune activating receptor is a CAR (e.g., anti-CD19-4-1BB-CD3 ζ CAR).

(78) In other embodiments, the single-chain variable fragment (scFv) linked to a localizing domain is selected from any one or more constructs shown in FIG. 2.

(79) In some aspects, the engineered immune cell is administered by infusion into the subject. Methods of infusing immune cells (e.g., allogeneic or autologous immune cells) are known in the art. A sufficient number of cells are administered to the recipient in order to ameliorate the symptoms of the disease. Typically, dosages of 10^{sup.7} to 10^{sup.10} cells are infused in a single setting, e.g., dosages of 10^{sup.9} cells. Infusions are administered either as a single 10^{sup.9} cell dose or divided into several 10^{sup.9} cell dosages. The frequency of infusions can be every 3 to 30 days or even longer intervals if desired or indicated. The quantity of infusions is generally at least 1 infusion per subject and preferably at least 3 infusions, as tolerated, or until the disease symptoms have been ameliorated. The cells can be infused intravenously at a rate of 50-250 ml/hr. Other suitable modes of administration include intra-arterial infusion, direct injection into tumor and/or perfusion of tumor bed after surgery, implantation at the tumor site in an artificial scaffold, intrathecal administration, and intraocular administration. Methods of adapting the present invention to such modes of delivery are readily available to one skilled in the art.

(80) In certain aspects, the cancer to be treated is a solid tumor or a hematologic malignancy. Examples of hematologic malignancies include acute myeloid leukemia, chronic myelogenous leukemia, myelodysplasia, acute lymphoblastic leukemia, chronic lymphocytic leukemia, multiple myeloma, Hodgkin and non-Hodgkin lymphoma. Examples of solid tumors include lung cancer, melanoma, breast cancer, prostate cancer, colon cancer, renal cell carcinoma, ovarian cancer, pancreatic cancer, hepatocellular carcinoma, neuroblastoma, rhabdomyosarcoma, brain tumor.

(81) In another embodiment, the present invention relates to a method for producing an engineered immune cell of the present invention, comprising introducing into an immune cell a nucleic acid comprising a nucleotide sequence encoding an immune activating receptor, and a nucleic acid comprising a nucleotide sequence encoding a target-binding molecule linked to a localizing domain, thereby producing an engineered immune cell.

(82) In certain embodiments, the nucleic acid comprising a nucleotide sequence is introduced into an immune cell *ex vivo*. In other embodiments, the nucleic acid comprising a nucleotide sequence is introduced into an immune cell *in vivo*.

(83) In some embodiments, an “immune cell” includes, e.g., a T cell, a natural killer (NK) cell, an NK/T cell, a monocyte, a macrophage, or a dendritic cell.

(84) The nucleic acid comprising a nucleotide sequence to be introduced can be a single bicistronic construct containing an immune activating receptor described herein and a target-binding molecule (e.g., scFv) linked to a localizing domain. As described herein, a single bicistronic construct can be prepared by inserting an internal ribosomal entry site (IRES) or a 2A peptide-coding region site between the 2 cDNAs encoding the immune activating receptor as described herein (e.g., CAR) and the target-binding molecule (e.g., scFv). The design of tricistronic delivery systems to delete more than one target should also be feasible. Alternatively, separate transductions (simultaneously or sequentially) of the individual constructs (e.g., CAR and LD-linked target-binding molecule) could be performed. Methods of introducing exogenous nucleic acids are exemplified herein, and are well-known in the art.

(85) As used herein, the indefinite articles “a” and “an” should be understood to mean “at least one” unless clearly indicated to the contrary.

EXEMPLIFICATION

Methods

(86) Cloning of scFv from Mouse Anti-Human CD3 Hybridoma

(87) PLU4 hybridoma cells secreting an anti-human CD3 monoclonal antibody (IgG2a isotype; Creative Diagnostics, Shirley, NY) were cultured in IMDM plus GlutaMAX medium (Life Technologies, Carlsbad, CA) with 20% fetal bovine serum (Thermo Fisher Scientific, Waltham, MA) and

antibiotics. Total RNA was extracted using TRIzol reagent (Life Technologies), and cDNA was synthesized using M-MLV reverse transcriptase (Promega, Madison, WI) and Oligo(dT).sub.15 primer (Promega). IgG Library Primer Set Mouse BioGenomics (US Biological, Salem, MA) was used to amplify the variable region of heavy chain (VH) and light chain (VL); PCR products were cloned into TOPO TA cloning kit for sequencing (Life Technologies). The VH and VL genes were assembled into scFv by a flexible linker sequence which encodes (Gly.sub.4Ser).sub.4 using splicing by overlapping extension-PCR. Signal peptide domain of CD8 α was subcloned by PCR using cDNA derived from human activated T cell from healthy donor, and connected to 5' end of the VL fragment. The Myc tag (EQKLISEEDL; SEQ ID NO: 1) was added to C-terminus of VH by PCR using sense primer: 5'-ATATATGAATTCGGCTTCCACCATGGCCTTACCAGTGACC-3' (SEQ ID NO: 2) and reverse primer: 5'-CAGATCTTCTTCAGAAATAAGTTTTTGTTCGGCTGAGGAGACTGTGAGAG-3' (SEQ ID NO: 3). Also the KDEL (SEQ ID NO: 4) coding sequence was generated after Myc tag by sense primer: 5'-ATATATGAATTCGGCTTCCACCATGGCCTTACCAGTGACC-3' (SEQ ID NO: 5) and reverse primer: 5'-TATATACTCGAGTTACAACCTCGTCCTTCAGATCTTCTTCAGAAATAAG-3' (SEQ ID NO: 6). The synthesized gene consisting of CD8 signal peptide, scFv against human CD3, Myc tag and KDEL (SEQ ID NO: 4) sequence was subcloned into EcoRI and XhoI sites of the MSCV-IRES-GFP vector. Constructs in which myc-KDEL was replaced by other sequences were also made as listed in FIG. 2.

(88) The sequence of "PEST"-SHGFPPEVEEQDDGTLPMSCAQESGMDRHPAACASARINV (SEQ ID NO: 7) motif corresponding to amino acids 422-461 of mouse ornithine decarboxylase was obtained from GenBank (accession number NM_013614.2). Codon optimization and gene synthesis was done by GenScript (Piscataway, NJ), and subcloned into the 3' end of VH by PCR. The constructs were subcloned into EcoRI and XhoI sites of the MSCV-IRES-GFP vector.

(89) Cloning of scFv Against Human CD7

(90) The sequence scFv derived from murine TH69 (anti-CD7) antibody was obtained from literature (Peipp et al., *Cancer Res* 2002 (62): 2848-2855). After codon optimization, the synthesized gene consisting of CD8 signal peptide, scFv against human CD7, Myc tag and KDEL (SEQ ID NO: 4) sequence was subcloned into EcoRI and XhoI sites of the MSCV-IRES-GFP vector. Constructs in which myc-KDEL was replaced by other sequences were also made as listed in FIG. 2.

(91) Cloning of scFv Against Human Beta-2 Microglobulin (hB2MG)

(92) The sequence scFv derived from murine BBM.1 (anti-hB2MG) IgG2b antibody was obtained from literature (Grosvender, E. A. et al., *Kidney Int.* 2004; 65(1):310-322). After codon optimization, synthesized gene consists of CD8 signal peptide, scFv against human B2MG, Myc tag and KDEL (SEQ ID NO: 4) sequence was subcloned into EcoRI and XhoI sites of the MSCV-IRES-GFP vector.

(93) Cloning of scFv against human KIR2DL1 and KIR2DL2/DL3

(94) The amino acid sequence of human monoclonal antibody I-7F9 (anti-KIR2DL1, KIR2DL2, and KIR2DL3) was derived from published International Patent Application WO2006003179 A2 by Moretta et al. After codon optimization, the sequence of scFv was designed by connecting variable light (VL) region and variable heavy (VH) region with linker sequence. The synthesized gene consisting of CD8 signal peptide, scFv against human KIRs (KIR2DL1, KIR2DL2 and KIR2DL3), CD8 hinge and transmembrane domain, and KKMP sequence was subcloned into EcoRI and XhoI sites of the MSCV-IRES-GFP vector. Constructs in which KKMP was replaced by other sequences were also made as listed in FIG. 2.

(95) Cloning of scFv Against Human NKG2A

(96) The sequence of murine antibody Z199 (anti-NKG2A) was derived from the published patent by Spee et al. (EP2247619 A1). After codon optimization, the sequence of scFv was designed by connecting variable light (VL) region and variable heavy (VH) region with linker sequence. The synthesized gene consisting of CD8 signal peptide, scFv against human NKG2A, CD8 hinge and transmembrane, and KKMP sequence was subcloned into EcoRI and XhoI sites of the MSCV-IRES-GFP vector. Constructs in which KKMP was replaced by other sequences were also made as listed in FIG. 2. The sequence information for the scFvs generated herein is shown in Table 1. Sequence information for the various components depicted in FIG. 2 is shown in Table 2.

(97) Anti-CD19-4-1BB-CD3 ζ CAR

(98) This CAR was generated as previously described (Imai, C. et al., *Leukemia*. 2004; 18:676-684; Imai, C. et al., *Blood*. 2005; 106:376-383).

(99) TABLE-US-00001

TABLE	1	scFv	sequence	information	Target	VH	amino acid	VL	amino acid	VH	cDNA	VL	cDNA	CD3																															
EVQL	QQSGAELAR	QIVLTQSPAIMSA	GAGGTCCAGCTGCAGCAG	CAAATTGTTCTCACCCAG	PGASVKMSCKAS	SPGKEVTMTCSAS	TCTGGGGCTGAACTGGCA	TCTCCAGCAATCATGTCT	GYTFTRYTMHWV	SSVS	SYMNWYQQ	AGACCTGGGGCCTCAGTG	GCATCTCCAGGGGAGAA	QKRPQGLEWIGY	KSGTSPKRWIYDT	AAGATGTCCTGCAAGGCTT	GGTCACCATGACCTGCA	INPSRGYTNYNQK	SKLASAGPAHFR	CTGGCTACACCTTTACTAG	GTGCCAGCTCAAGTGTA	FKDKATLTITDKSS	GS	SGTSYS	SLTIS																				
TACT	CACGACTGGGT	AGTTACATGAAC	GGTAC	STAYMQLSSLTSE	GMEAEADAATYYC	AAAACAGAGCGCTGGACA	CAGCAGAAGTCAGGCAC	DSAVYYCARYYD	QQWSSNPFTFGSG	GGGTCTGGAATGGATTGG	CTCCCCCAAAAGATGGA	DHYCLDYWGQGT	TKLEINR	(SEQ ID ATACATTAATCCTAGCCGT	TTTATGACACATCCAAAC	TLTVSSA	(SEQ ID NO: 13)	GGTTATACTAATTACAATC	TGGCTTCTGGAGTCCCTG	NO: 12)	AGAAGTTCAAGGACAAGG	CTCACTTCAGGGGCGAGTG	CCACATTGACTACAGACA	GGTCTGGGACCTCTTACT	AATCCTCCAGCACAGCCTA	CTCTCACAATCAGCGGCA																			
CATG	CAACTGAGCAGCCT	TGGAGGCTGAAGATGCT	GACATCTGAGGACTCTGCA	GCCACTTATTACTGCCAG	GTCTATTACTGTGCAAGAT	CAGTGGAGTAGTAACCC	ATTATGATGATCATTACTG	ATTCACGTTCCGGCTCGGG	CCTTGACTACTGGGCGCAA	GACAAAGTTGGAATAA	GGCACCTCTCACAGTCT	ACCGG	(SEQ ID NO: 15)	14)	CD7	EVQLVESGGGLVK	AAYKDIQMTQTT	GAGGTGCAGCTGGTTCGAA	GCCGCATACAAGGATAT	(TH69)	PGGSLKLSCAASG	SSLSASLGDRVTIS	TCTGGAGGAGGACTGGTG	TCAGATGACTCAGACCA	LTFSSYAMSWVR	CSASQGISNYLN																			
AAGCC	CAGGAGGATCTCTG	CAAGCTCCCTGAGCGCCT	QTPEKRLEWVASI	WYQKQPDGTVKL	AAACTGAGTTGTGCCGCTT	CCCTGGGAGACCGAGTG	SSGFTYYPDSVK	LIYTTSSLHSGVP	CAGGCCTGACCTTCTCAAG	ACAATCTCTTGCACTGCA	GRFTISRDNARNIL	SRFSGSGSGTDYS	CTACGCCATGAGCTGGGTG	TCACAGGGAATTAGCAA	YLMSSLRSEDTA	LTISNLEPEDIATY	CGACAGACACCTGAGAAG	CTACCTGAATTGGTATCA	MYCARDEVGRY	YCQYQSKLPYTF	CGGCTGGAATGGGTGCGT	GCAGAAGCCAGATGGCA	LDVWVGAGTTVT	GGG	TKLEIKR	AGCATCTCCTCTGGCGGGT	CTGTGAAAGCTGTGATCT	SS	(SEQ ID NO: 16)																
(SEQ ID NO: 17)	TCACATACTACGAGCT	ACTACTACTCTAGTCTGC	CGTGAAGAGCGAGATTACT	ACAGTGGGGTCCCTCAC	ATCTCTCGGGATAACGCAA	GATTCAGCGGATCCGGCT	GAAATATTCTGTACCTGCA	CTGGGACAGACTACAGC	GATGAGTTCAGTGGAG	CTGACTATCTCCAACCTG	CGAGGACACCGCAATGTA	GAGCCCGAAGATATTGC	CTATTGTGCCAGGGACGA	CACCTACTATTGCCAGCA	AGTGCGCGGCTATCTGGAT	GTACTCCAAGCTGCCTTA	GTCTGGGGAGCTGGCACT	TACCTTTGGCGGGGGAA	ACCGTCACCGTCTCCAGC	CAAAGCTGGAGATTA	AAA	(SEQ ID NO: 18)	AGG	(SEQ ID NO: 19)	CD7	QVQLQESGAELVK	DIELTQSPATLSVT	CAGGTCCAGCTGCAGGAG	GACATCGAGCTGACACA	(3a1f)	PGASVKLSCKASG	PGDSVLS	SCRASQ	TCAGGGGCAGAGCTGGTG	GTCTCCAGCCACTCTGAG	YTFTSYWMHWVK	SISNNLHWYQK	AAACCCGAGCGAGTGTC	CGTGACCCCTGGCGGATTC	QRPGQLEWIGKI	SHEPRLLIKSASQ	AAACTGTCTGTGAAGGCA	TGTCACTGTCTCATGTAG	NPNGRNTYNEKF	SISGIPSRFSGSGS
GCGGCTATACTTTCCACAG	AGCTAGCCAGTCCATCTC	KSKATLTVDKSS	GTDFTL	SINSVETE	CTACTGGATGCACTGGGTG	TAACAATCTGCACTGGTA	TAYMQLSSLTSED	DFGMYFCQQSNS	AAACAGAGGCCAGGACAG	CCAGCAGAAATCACATG	SAVYYCARGGVY	WPYTFGGG	TKLEI	GGCCTGGAGTGGATCGGC	AAAGCCCTCGGCTGCTG	YDLYYYALDYGW	KR	(SEQ ID NO: AAGATTAACCCAGCAAT	ATTAAGAGTGCTTCACAG	QGTTVT	VSS	(SEQ 21)	GGGCGCACCAACTACAAC	AGCATCTCCGGGATTCCA	ID NO: 20)	GAAAAGTTTAAATCCAAG	AGCAGATTCTCTGGCAGT	GCTACACTGACTGTGGACA	GGGTCAGGAACCGACTT	AGAGCTCCTTACCGCATA	TACACTGTCCATTA	ACTC	CATGCAGCTGAGTTCAGT												

TGTGGAGACCGAAGATT ACATCTGAAGATGATGTC TCGGATGATGATTTTTGCC GTGTACTATTGTCGCCGAGG
AGCAGAGCAATTCCTGG GCGGGGTCTACTATGACCT CTTACACATTCGGAGGC GTACTATTACGCACTGGAT
GGGACTAAACTGGAGAT TATTGGGGGCAGGGAACC TAAGAGG (SEQ ID NO: ACAGTGACTGTCAGCTCC 23)(SEQ ID NO:
22) CD45 QVQLVESGGGLV DIVLTQSPASLAV CAGGTGCAGCTGGTGCAG GACATTGTGCTGACCCAG QPGGSLKLSCAAS
SLGQRTTISCRAS TCTGGAGGAGGACTGGTG TCCCCTGCTTCACTGGCA GFDFSRYWMSWV KSVSTSGYSYLH
CAGCCTGAGGAAGTCTGT GTGAGCCTGGGACAGAG RQAPGKGLEWIGE WYQQKPGQPPKL AAGCTGTCTATGTCAGCCA
GGCAACCATCAGCTGCC INPTSSSTINFPSLK LIYLASNLESVGP GCGGGTTCGACTTTTCTCG GAGCCTCTAAGAGTGTCT
DKVFISRDNAKNT ARFSGSGSGTDFT ATACTGGATGAGTTGGGTG CAACAAGCGGATACTCC LYLQMSKVRSED LNIHPVEEEDAAT
CGGCAGGCACCAGGAAAA TATCTGCACTGGTACCAG ALYYCARGNYR YYCQHSRELPTF GGAAGTGAATGGATCGGC
CAGAAGCCAGGACAGCC YGDAMDYWGQG GSGTKLEIK (SEQ GAGATTAACCCAACTAGCT ACCTAAACTGCTGATCTA TSVTVS
(SEQ ID ID NO: 25) CCACCATCAATTTACACC TCTGGCTTCCAACCTGGA NO: 24) CAGCCTGAAGGACAAAGT
ATCTGGAGTGCCTGCACG GTTTATTTCCAGAGATAAC CTTCTCCGATCTGGAAG GCCAAGAATACTCTGTATC
TGAACACCGATTTACACT TGCAGATGTCCAAAGTCA GAATATTCACCCAGTCGA GGTCTGAAGATACCGCCCT
TGAAGAGGATGCCGCTA GTACTATTGTGCTCGGGC CTAATAATTGCCAGCACA AACTACTATAGATACCGG
GCCGGGAGCTGCCCTTCA GACGCTATGGATTATTGGG CATTTGGCAGCGGGACT GGCAGGGAAGTACGCTGA
AAGCTGGAGATCAAG CCGTGAGT (SEQ ID NO: (SEQ ID NO: 27) 26) B2MG EVQLQQSGAELVK DIQMTQSPASQSA
GAGGTGCAGCTGCAGCAG GATATTCAGATGACCCA PGASVKLSCTPSG SLGESVTITCLAS AGCGGAGCAGAACTGGTG
GTCCCCTGCATCACAGAG FNVKDTYIHWVK QTIGTWLAWYQQ AAACCTGGAGCCAGCGTC CGCCTCCCTGGGCGAGTC
QRPKQGLEWIGRI KPGKSPQLLIYAA AAGCTGTCTGTACTCCAT AGTGACCATCACATGCCT DPSDGDIKYDPKF TSLADGVPSRFSG
CTGGCTTCAACGTGAAGG GGCTAGCCAGACAATTG QGKATITADTSSN SGSGTKFSLKIRT ACACATACATTCAGTGGGT
GAGTCTGGCTGGCTATGGT TVSLQLSLTSED LQAEDFVSYQC CAAGCAGCGGCCCCAAACA ACCAGCAGAAGCCCCGGC
AVYYCARWFGDY QLYSKPYTFGG GGGACTGGATGGATCGG AAATCCCCTCAGTGGCTG GAMNYWGQTSV TKLEIKRAD (SEQ
CAGAATTGACCCATCCGAC ATCTATGCAGCTACCTCT TVSS (SEQ ID NO: ID NO: 29) GGCGATATCAAGTATGATC
CTGGCAGACGGAGTGCC 28) CCAAATTCAGGGGAAGG CAGTAGGTTCTCTGGGAG CTACTATTACCGCAGATAC
TGGATCAGGCACCAAGT CAGCTCCAACACAGTGAG TTTCTCTGAAAATTGCA TCTGCAGCTGTCTAGTCTG
CACTGCAGGCTGAGGAT ACTAGCGAAGACACCGCC TTCGTCTCCTACTATTGC GTCTACTATTGTGCTAGAT
CAGCAGCTGTACTCTAAA GGTTCGGCGATTACGGGGC CTTATACATTTGGCGGG CATGAATTATTGGGGGCA
GGAATAEAAGCTGGAAAT GGAACACAGCGTCACCGT CAAACGAGCAGAC (SEQ GTCCAGC (SEQ ID NO: ID NO: 31) 30)
NKG2A EVQLVESGGGLVK QIVLTQSPALMSA GAGGTGCAGCTGGTGGAG CAGATTGTCTGACCCAG PGGSLKLSCAASG
SPGEKVTMTCSAS AGCGGAGGAGGACTGGTG TCTCCAGCCCTGATGAGC FTFSSYAMSWVRQ SSVSYIYWYQK
AAGCCAGGAGGAAGCCTG GCCTCCCCTGGCGAGAA SPEKRLEWVAEISS PRSSPKWIYLS AAGCTGTCTGTGCCGCCT
GGTGACAATGACCTGCTC GGSYTYYPDVTG NLASGVPARFSGS CTGGCTTCACATTTTCTC TGCCAGCTCCTCTGTGAG
RFTISRDNANKNTL GSGTSYSLTISSM TTATGCAATGAGCTGGGTG CTACATCTATTGGTACCA YLEISSLRSED TAM EAEDAATYYCQK
CGGCAGTCCCCAGAGAAG GCAGAAGCCTCGGAGCT YYCTRHDYPRFF WSGNPYTFGGGT AGACTGGAGTGGGTGGCA
CCCCAAAGCCCTGGATCT DVWGAGTTVTVS KLEIKR (SEQ ID GAGATCAGCTCCGGAGGA ATCTGACATCCAACCTGG S (SEQ
ID NO: 32) NO: 33) TCCTACACCTACTATCCTG CCTCTGGCGTGCCAGCCA ACACAGTGACCGGCCGGT
GATTCTCTGGCAGCGGT TCACAATCTCTAGAGATAA CCGGCACATCTTACAGCC CGCCAAGAATACCTGTAT
TGACCATCTCTAGCATGG CTGGAGATCTCTAGCTGA AGGCCGAGGACGCCGCC GATCCGAGGATACAGCCA
ACCTACTATTGCCAGCAG TGTACTATTGCACCAGGCA TGGTCCGGCAATCCATAT CGGCGACTACCCACGCTTC
ACATTTGGCGGCGGCAC TTTGACGTGTGGGGAGCA CAAGCTGGAGATCAAGA GGAACCACAGTGACCGTG GG (SEQ ID NO:
35) TCCTCT (SEQ ID NO: 34) KIR QVQLVQSGAEVK EIVLTQSPVTL SLS CAGGTCCAGCTGGTGCAGT GAGATCGTGTGACCCA
2DL1 KPGSSVKVSKAS PGERATLSCRASQ CTGGAGCTGAAGTGAAGA GTCTCCTGTCACTGAG and 2/3 GGTF SFYAI SWVR
SVSSYLAWYQK AACCAGGGAGCTCCGTCA TCTGTCAACAGGGGAAC QAPGQGLEWMGG PGQAPRLIYDAS
AGGTGTCATGCAAAGCAA GGGCTACACTGTCTTGCA FIPIFGAANYAQKF NRATGIPARFSGS GCGGCGGACTTTCTCCTT
GAGCAAGCCAGTCCGTG QGRVTITADEST TSGTDFLT LTISSLE TTATGCAATCTCTTGGGTG AGTCTCATCTGGCCTGG
AYMELSSLRSDDT PEDFAVYYCQR AGACAGGCACCTGGAGCAG TATCAGCAGAAGCCAGG AVYYCARIPSGSY SNWMTYTFGQGTK
GGACTGGAGTGGATGGGA CCAGGCTCCAGGCTGCT YYDYDMDVWGQ LEIKRT (SEQ ID GGCTTCATCCCAATTTTGT
GATCTACGATGCAAGCA GTTVTVSS (SEQ NO: 37) GAGCCGCTAACTATGCCCA ACAGGGCCACTGGGATT ID NO: 36)
GAAGTTCAGGGCAGGGT CCCGCCGCTTCTCTGGC GACCATCACAGCTGATGA AGTGGGTGAGGAACCGA
GTCTACTAGTACCGCATAC CTTACTCTGACCATTTT ATGGAAGTGTCTAGTCTGA TAGTCTGGAGCCTGAAG
GGAGCGACGATACCGCCG ATTTGCGCGTGACTATT TGTACTATTGTGCTCGCAT GCCAGCAGCGATCCAAT
TCCATCAGGCAGTACTAT TGGATGTATACTTTTGGC TACGACTATGATATGGACG CAGGGGACCAAGCTGGA
TGTGGGGCCAGGGGACCA GATCAAACGGACA (SEQ CAGTCACCGTGAGCAGC ID NO: 39) (SEQ ID NO: 38)
(100) TABLE-US-00002 TABLE 2 Sequence information for components depicted in FIG. 2 CD8 SP VH-VL amino linker CD8
hinge and VH-VL linker CD8 hinge and TM Target acid amino acid TM amino acid CD8 SP cDNA cDNA cDNA CD3 MALPVT
GGGSGG KPTTTPAPRPPTP ATGGCCTTACC GGTGGTGGTG AAGCCCACCACG ALLLPLA GSGGGG APTIASQPLSLRP
AGTGACCGCCT GTTCTGGTGG ACGCCAGCGCCG LLLHAAR SGGGS EACRPAAGGAVH TGCTCCTGCCG TGGTGGTTCT
CGACCACCAACA P (SEQ ID (SEQ ID TRGLDFACDIYI CTGGCCTTGCT GCGGCGCGCG CCGGCGCCACC NO: 40) NO: 41)
WAPLAGTCGVLL GCTCCACGCCG GCTCCGGTGG ATCGCGTCGCAG LSLVITLY (SEQ CCAGGCCG TGGTGGATCC
CCCCTGTCCCTGC ID NO: 42) (SEQ ID NO: (SEQ ID NO: GCGGAGGCGT 44) 51) GCGGCGCAGCGG CGGGGGCGCAG
TGCACACGAGGG GGCTGGACTTCG CCGTGATATCTA CATCTGGGCGCC CTTGGCGGAC TTGTGGGTCTCT CTCTGTCACTGG
TTATCACCTTTA C (SEQ ID NO: 57) CD7 MALPVT GGGSGG TTPAPRPPTPAP ATGGCTCTGCC GGAGGAGGAG
ACCACTACACCT (TH69) ALLLPLA GSGGGG TIASQPLSLRPEA TGTGACCGCAC GAAGCGGAGG GCACCAAGGCCT LLLHAAR
SGGGS CRPAAGGAVHTR TGCTGTGCCC AGGAGGATCC CCCACACCCGCTC P (SEQ ID (SEQ ID GLDFACDIYIWA
CTGGCTCTGCT GGAGGCGGGG CCACTATCGCTTC NO: 40) NO: 41) PLAGTCGVLLS GCTGCACGCCG GATCTGGAGG
CCAGCCACTGTCC LVITLY (SEQ ID CAAGACCT AGGAGGAAGT CTGAGGCCGAG NO: 50) (SEQ ID NO: (SEQ ID NO:
GCCTGCAGGCCA 45) 52) GCAGCTGGCGGA GCCGTGCATACT AGGGGGCTGGAC TTCGCTTGCGACA TCTACATCTGGG
CCCCTGGPCAGG GAGCTGCGGAGT CCGTGTGCTGTCC CTGGTCATCACAC TTAC (SEQ ID NO: 58) CD7 MALPVT GGGSGG
TTPAPRPPTPAP ATGACTCTGCC GGAGGAGGAG ACTACCACCA (3a1f) ALLLPLA GSGGGG TIASQPLSLRPEA CGTACCGCTC
GATCCGCGCG GCTCCAAGACCA LLLHAAR S (SEQ ID CRPAAGGAVHTR TGCTGTGCTCT AGGAGGCTCT CCTACCCCTGCAC P
(SEQ ID NO: 43) GLDFACDIYIWA CTGGCTCTGCT GGGGGAGGCG CAACAATTGCTA NO: 40) PLAGTCGVLLS GCTGCACGCTG
GGAGT (SEQ GTCAGCCACTGTC LVITLY (SEQ ID CTCGACCA ID NO: 53) ACTGAGACCAGA NO: 50) (SEQ ID NO:
AGCATGTAGGCC 46) TGCAGCTGGAGG AGCTGTGCACAC CAGAGGCCTGGA CTTTGCCTGCGAT ATCTACATTTGGG
CTCCTCTGGCAG AACCTGTGGCGT GCTGCTGCTGTCT CTGGTCATCACAC TTAC (SEQ ID NO: 59) CD45 MALPVT
GGGSGG KPTTTPAPRPPTP ATGGCTCTGCC GGAGGAGGAG AAGCCCACCACG ALLLPLA GSGGGG APTIASQPLSLRP

CGTAGCAGGCTC GAAGTGAGG ACGGACGCGCCG LLHAAR SGGGGS EACRPAAGGAVH TGCTGCTGCTT AGGAGATCA
CGACCACCAACA P (SEQ ID (SEQ ID TRGLDFACDIYI CTGGCTCTGCT GGAGGCGGGG CCGGCGCCCACC NO: 40) NO: 41)
WAPLAGTCGVLL GCTGCATGCTG GAAGCGGCGG ATCGCGTCGCAG LSLVITLY (SEQ CTCGACCT GGGAGGCTCC
CCCCTGTCCCTGC ID NO: 42) (SEQ ID NO: (SEQ ID NO: GCCCAGAGGCGT 47) 54) GCCGGCCAGCGG CGGGGGGCGCAG
TGCACACGAGGG GGCTGGACTTCG CCTGTGATATCTA CATCTGGGCGCC CTTGGCCGGGAC TTGTGGGGTCCTT CTCCTGTCACTGG
TTATCACCCCTTTA C (SEQ ID NO: 57) B2MG MALPVT GGGGSGG KPTTTPAPRPPTP ATGGCCCTGCC GGAGGAGGAG
AAGCCACACG ALLLPLA GSGGGG APTIASQPLSLRP CGTACCGCCCC GAAGTGAGG ACGCCAGCGCCG LLHAAR SGGGGS
EACRPAAGGAVH TGCTGCTGCC AGGAGGGTCA CGACCACCAACA P (SEQ ID (SEQ ID TRGLDFACDIYI CTGGCTCTGCT
GGAGGCGGGG CCGGCGCCCACC NO: 40) NO: 41) WAPLAGTCGVLL GCTGCACGCCG GAAGCGGCGG ATCGCGTCGCAG
LSLVITLY (SEQ CAAGACCC GGGAGGATCC CCCCTGTCCCTGC ID NO: 42) (SEQ ID NO: (SEQ ID NO: GCCCAGAGGCGT
48) 55) GCCGGCCAGCGG CGGGGGGCGCAG TGCACACGAGGG GGCTGGACTTCG CCTGTGATATCTA CATCTGGGCGCC
CTTGGCCGGGAC TTGTGGGGTCCTT CTCCTGTCACTGG TTATCACCCCTTTA C (SEQ ID NO: 57) NKG2A MALPVT GGGGSGG
KPTTTPAPRPPTP ATGGCTCTGCC GGAGGAGGAG AAGCCAACCACA LLPLA GSGGGG APTIASQPLSLRP CGTGACCGCCC
GATCTGGAGG ACCCTGCACCA LLHAAR SGGGGS EACRPAAGGAVH TGCTGCTGCC AGGAGGCGG AGGCCACCTACA P
(SEQ ID (SEQ ID TRGLDFACDIYI CTGGCTCTGCT GCGGCGGCGG CCAGCACCTACC NO: 40) NO: 41) WAPLAGTCGVLL
GCTGCACGCTG GCTCCGGCGG ATCGCAAGCCAG LSLVITLY (SEQ CCCGCCA CGGCGGCTCT CCACTGTCCCTGA ID NO: 42)
(SEQ ID NO: (SEQ ID NO: GGCCAGAGGCAT 49) 56) GTAGGCCTGCAG CAGGAGGCGCCG TGCACACACGCG
GCCTGGACTTTGC CTGCGATATCTAC ATCTGGGCACCA CTGGCAGGAACC TGTGGCGTGCTGC TGCTGAGCCTGGT
GATTACCCTGTAT (SEQ ID NO: 60) KIR MALPVT GGGGSGG KPTTTPAPRPPTP ATGGCCTTACC GGTGGTGGTG AAGCCCACCAG
2DL1 ALLLPLA GSGGGG APTIASQPLSLRP AGTGACCGCCT GTTCTGGTGG ACGCCAGCGCCG and 2/3 LLHAAR SGGGGS
EACRPAAGGAVH TGCTCCTGCCG TGGTGGTTCT CGACCACCAACA P (SEQ ID (SEQ ID TRGLDFACDIYI CIGGCcrTGur
GGCGGCGCGG CCGGCGCCCACC NO: 40) NO: 41) WAPLAGTCGVLL GTCCACGCCG GCTCCGGTGG ATCGCGTCGCAG
LSLVITLY (SEQ CCAGGCGG TGGTGGATCC CCCCTGTCCCTGC ID NO: 42) (SEQ ID NO: (SEQ ID NO: GCCCAGAGGCGT
44) 51) GCCGGCCAGCGG CGGGGGGCGCAG TGCACACGAGGG GGCTGGACTTCG CCTGTGATATCTA CATCTGGGCGCC
CTTGGCCGGGAC TTGTGGGGTCCTT CTCCTGTCACTGG TTATCACCCCTTTA C (SEQ ID NO: 57)

Gene Transduction, Cell Expansion, Flow Cytometric Analysis and Functional Studies

(101) These were performed as previously described (Kudo, K et al., *Cancer Res.* 2014; 74(1):93-103).

Results

(102) Generation of scFv Constructs

(103) A schematic of the technology is outlined in FIG. 1. A schematic representation of the inhibitory constructs that we generated is shown in FIG. 2. The scFv portion can be derived by cloning the cDNA encoding variable light (VL) and variable heavy (VH) immunoglobulin chain regions from an antibody-producing hybridoma cell line or from the corresponding published sequences. VL and VH are linked with a short peptide sequence ("linker") according to standard techniques to make a full scFv. To be expressed, the scFv is linked to a signal peptide at the N-terminus; the signal peptide is required for the scFv to be expressed, as confirmed in preliminary experiments. Proteins containing scFv plus signal peptide are generally released into the cells' milieu. For example, in preliminary experiments (not shown), an anti-CD3 ϵ scFv plus signal peptide expressed in Jurkat T cells was detected in the cells' culture supernatant. By directing the scFv to specific compartments and preventing its secretion, possible effects on other cells are prevented. To direct it to the endoplasmic reticulum (ER), the KDEL (SEQ ID NO: 4) motif (which retains proteins in the ER) was utilized (Strebe N. et al., *J Immunol Methods.* 2009; 341(1-2):30-40). To promote the degradation of the targeted protein, we linked it to a proteasome-targeting PEST motif (Joshi, S. N. et al., *MAbs.* 2012; 4(6):686-693). The scFv can also be directed to the cell membrane by linking it to the transmembrane domain and hinge of CD8 α or another transmembrane protein.

(104) Downregulation of T-Cell Receptor in T Lymphocytes Expressing Anti-CD19-BB- ζ CAR

(105) To determine whether the proposed strategy could be applied to generate immune cells expressing CAR and lacking one or more markers, T-cell receptor (TCR) expression was downregulated in anti-CD19 CAR T-cells.

(106) To be expressed on the cell membrane, the CD3/TCR complex requires assembly of all its components (TCR α , TCR β , CD3 δ , CD3 ϵ , CD3 γ , CD3 ζ). Lack of one component prevents CD3/TCR expression and, therefore, antigen recognition. In preliminary studies, the scFv from an anti-CD3 ϵ hybridoma (purchased from Creative Diagnostics, Shirley, NY) was cloned and generated the constructs containing KDEL (SEQ ID NO: 4), PEST, CD8 α transmembrane domain or others as shown in FIG. 2.

(107) The constructs disclosed herein were transduced in the CD3/TCR+ Jurkat cell line using a murine stem cell virus (MSCV) retroviral vector containing green fluorescent protein (GFP). Percentage of GFP+ cells after transduction was >90% in all experiments. FIG. 3A shows results of staining with anti-CD3 ϵ antibody among GFP+ cells, as measured by flow cytometry. Antibody staining of CD3 ϵ was decreased to variable degree in cells transduced with the constructs listed. Similar downregulation of CD3 ϵ was obtained with human peripheral blood T lymphocytes (FIG. 3B). FIG. 3C shows illustrative flow cytometry dot plots of CD3 ϵ expression in GFP-positive Jurkat cells after transduction with different gene constructs in comparison with cells transduced with a vector containing GFP alone. Downregulation of CD3 did not affect growth of Jurkat cells or expression of all other cell markers tested, including CD2, CD4, CD8, CD45, CD25, CD69. Inhibition of CD3 expression persisted for over 3 months. Further enrichment of CD3-negative cells could be achieved by CD3+ T cell depletion with anti-CD3 magnetic beads (Dynal, Life Technologies, Carlsbad, CA).

(108) Staining with anti-TCR $\alpha\beta$ antibody of Jurkat cells or human peripheral blood T lymphocytes showed that down regulation of CD3 ϵ expression was associated with downregulation of TCR $\alpha\beta$ expression (FIG. 4).

(109) Next, it was determined whether the anti-CD3 scFv-myc KDEL could be expressed simultaneously with an anti-CD19-4-1BB-CD3 ζ CAR. As shown in FIG. 5, this resulted in T cells lacking CD3 expression while expressing the anti-CD19 CAR. TCR was also absent on these cells (not shown).

(110) To assess whether CAR could signal in Jurkat cells with downregulated CD3/TCR, the expression of the activation markers CD69 and CD25 was tested, and exocytosis of lytic granules was measured by CD107a expression in Jurkat cells co-cultured with the CD19+ leukemia cell line OP-1. As shown in FIG. 6, downregulation of CD3/TCR with the anti-CD3 scFv-myc KDEL construct did not diminish the capacity of anti-CD19-4-1BB-CD3 ζ CAR to activate Jurkat cells. To further explore the effects of CD3/TCR deletion on CAR signaling, it was determined whether CD3-negative T lymphocytes expressing the CAR could be stimulated by its ligation. As shown in FIG. 7, co-culture of T lymphocytes expressing the anti-CD19 CAR with CD19+ leukemic cells led to T cell proliferation regardless of whether CD3 was downregulated or not, indicating that CD3/TCR downregulation did not diminish the CAR proliferative stimulus.

(111) Accordingly, CD3/TCR can be effectively downregulated in CAR-T cells using the anti-CD3 scFv-myc KDEL construct without affecting T cell activation, degranulation and proliferation driven by the CAR.

(112) Downregulation of CD7

(113) It was determined whether the strategy that successfully modulated CD3/TCR expression could be applied to other surface molecules. For this purpose, CD7 expression was modulated. The scFv sequence was derived from that published by Peipp et al. (*Cancer Res* 2002 (62): 2848-2855), which was linked to the CD8 signal peptide and the myc-KDEL sequence as illustrated in FIG. 2. Using the MSCV retroviral vector, the anti-CD7-myc KDEL construct was transduced in peripheral blood lymphocytes, which have high expression of CD7 as detected by an anti-CD7 antibody

conjugated to phycoerythrin (BD Bioscience). As shown in FIG. 8, CD7 in T lymphocytes transduced with the construct was virtually abrogated.

(114) Downregulation of HLA-Class I

(115) The strategy was then applied to downregulate another surface molecule, HLA class I.

(116) HLA class I consists of polymorphic α chains and a non-polymorphic chain termed β 2-microglobulin. Knock-down of the latter subunit results in abrogation of HLA (MHC in the mouse) Class I expression (Koller, B H et al., *Science*. 1990; 248(4960):1227-1230). An scFv reacting with β 2-microglobulin was used to suppress expression of HLA Class I in immune cells.

(117) The scFv sequence was derived from that published by Grovender et al. (*Kidney Int.* 2004; 65(1):310-322), which was linked to the CD8 signal peptide and the myc KDEL sequence as illustrated in FIG. 2. Using the MSCV retroviral vector, the anti- β 2M-myc KDEL construct was transduced in Jurkat cells, which have high expression of HLA Class I as detected by an anti-HLA-ABC antibody conjugated to phycoerythrin (BD Pharmingen). As shown in FIG. 9, Jurkat cells transduced with the construct had a substantial downregulation of HLA-ABC expression. Cells maintained their morphology and growth capacity.

(118) Downregulation of Inhibitory Receptors in NK Cells

(119) To determine if the strategy outlined above would also apply to surface molecules expressed in other immune cells, downregulation of function of the inhibitory receptor KIR2DL1, KIR2DL2/DL3 and NKG2A was tested in NK cells.

(120) To downregulate KIR receptors, an scFv reacting with KIR2DL1 and KIR2DL2/DL3 was used to suppress their expression in NK cells. The scFv sequence was derived from that published by Moretta et al. (patent WO2006003179 A2), which was linked to the CD8 signal peptide and the ER retention sequences as illustrated in FIG. 2. Using the MSCV retroviral vector, the constructs were transduced in NK cells expanded from human peripheral blood and selected for KIR2DL1 expression. These cells had high KIR2DL1 expression as detected by an anti-KIR2DL1 antibody conjugated to allophycocyanin (R&D Systems) and also high KIR2DL2/DL3 expression as detected by an anti-KIR2DL2/DL3 antibody conjugated to phycoerythrin (BD Bioscience). FIG. 10 shows results obtained with scFv-linker(20) AEKEDL and scFv-EEKKMP, with substantial down regulation of the targeted KIRs.

(121) To downregulate NKG2A, an scFv reacting with NKG2A was used to suppress its expression in NK cells. The scFv sequence, which was derived from published European Patent Application No. EP2247619 A1 by Spee et al. was linked to the CD8 signal peptide and the ER retention sequences as illustrated in FIG. 2. Using the MSCV retroviral vector, the constructs were transduced in NK cells expanded from human peripheral blood, which had high NKG2A expression as detected by an anti-NKG2A antibody conjugated to phycoerythrin (Beckman Coulter). FIG. 11 shows substantial downregulation of NKG2A obtained with scFv-EEKKMP.

(122) The teachings of all patents, published applications and references cited herein are incorporated by reference in their entirety.

(123) While this invention has been particularly shown and described with references to example embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

Claims

1. An in vitro method for producing an engineered immune cell, the method comprising: introducing into an immune cell (i) a first polynucleotide sequence encoding an immune activating receptor comprising a first binding domain that binds to a molecule expressed on the surface of a cancer cell, and (ii) a second polynucleotide sequence encoding a target-binding molecule linked to a localizing domain, wherein the target-binding molecule comprises a second binding domain that binds to a target of the immune cell, and wherein the target-binding molecule linked to the localizing domain downregulates expression of the target of the immune cell, thereby producing an engineered immune cell, wherein the engineered immune cell retains similar or higher (i) expression of the immune activating receptor, (ii) expression of a T cell activation marker, (iii) T cell proliferation, or (iv) any combination thereof, as compared to that of an otherwise identical immune cell that comprises the first polynucleotide sequence but not the second polynucleotide sequence, wherein the second polynucleotide sequence comprises, from 5' terminus to 3' terminus, a first nucleotide sequence encoding the second binding domain, a second nucleotide sequence encoding a peptide linker, and a third nucleotide sequence encoding the localizing domain, and wherein the peptide linker comprises at least 5 amino acids, and wherein the molecule expressed on the surface of the cancer cell and the target of the immune cell are the same.
2. The in vitro method of claim 1, wherein the localizing domain 3 comprises an ER retention sequence, a Golgi retention sequence, or a PEST sequence.
3. The in vitro method of claim 1, wherein the first polynucleotide sequence and the second polynucleotide sequence are on the same polynucleotide molecule.
4. The in vitro method of claim 3, further comprising an internal ribosomal entry site or a 2A peptide-coding region site between the first polynucleotide sequence encoding the immune activating receptor and the second polynucleotide sequence encoding the target-binding molecule linked to a localizing domain.
5. The in vitro method of claim 2, wherein the localizing domain comprises an ER retention sequence.
6. The in vitro method of claim 5, wherein the ER retention sequence comprises a KDEL amino acid sequence.
7. The in vitro method of claim 5, wherein the ER retention sequence comprises a KKXX amino acid sequence, wherein X represents any amino acid.
8. The in vitro method of claim 2, wherein the localizing domain comprises a Golgi retention sequence.
9. The in vitro method of claim 8, wherein the Golgi retention sequence comprises a YQRL amino acid sequence.
10. The in vitro method of claim 2, wherein the localizing domain comprises a PEST sequence.
11. The in vitro method of claim 1, wherein the immune cell is a T cell.
12. The in vitro method of claim 1, wherein the immune cell is a natural killer (NK) cell.
13. The in vitro method of claim 1, wherein the second binding domain is a single chain binding domain.
14. The in vitro method of claim 13, wherein the single chain binding domain is a single chain variable fragment (scFv).
15. The in vitro method of claim 1, wherein the target-binding molecule linked to a localizing domain further comprises a transmembrane domain.
16. The in vitro method of claim 15, wherein the transmembrane domain is a CD8 transmembrane domain.
17. The in vitro method of claim 1, wherein the immune activating receptor is a chimeric antigen receptor (CAR) comprising the first binding domain, a signaling domain, and a costimulatory domain.
18. The in vitro method of claim 1, wherein the first binding domain and the second binding domain have the same sequence.
19. The in vitro method of claim 1, wherein the second binding domain and the first binding domain are different.
20. The in vitro method of claim 17, wherein the signaling domain comprises a CD3 ζ , a Fc, a DAP10, or a DAP12 domain.
21. The in vitro method of claim 17, wherein the costimulatory domain comprises a 4-1BB domain, a CD28 domain, a CD28.sub.LL.fwdarw.GG variant domain, an OX40 domain, an ICOS domain, a CD27 domain, a GITR domain, a HVEM domain, a TIM1 domain, a LFA1 domain, or a CD2 domain.
22. The in vitro method of claim 1, wherein the second polynucleotide sequence further comprises a nucleotide sequence encoding a signal peptide.

23. The in vitro method of claim 1, wherein the first binding domain is a first single chain variable fragment (scFv) and the second binding domain is a second scFv.
